



Optimal scheduling and motion planning of automated vehicles at intersections

Federico Gallo ^a,* , Alessandra Miagostovich ^a, Davide Giglio ^a, Angela Di Febbraro ^a,
Nicola Sacco ^a

^a Department of Mechanical, Energy, Management and Transportation Engineering, University of Genoa, Via Montallegro 1, Genoa, 16145, Italy

ARTICLE INFO

Keywords:

Automated vehicles
Optimal vehicle motion planning
Optimal vehicle scheduling
Autonomous intersection management

ABSTRACT

The ever-growing diffusion of automation in road transport and the spreading of communication technologies applied to road infrastructures toward so-called *smart roads* is leading to a need for coordination methods for automated vehicles, to fully exploit the potentialities of such technologies to make road transport more efficient, safer, and greener.

This study focuses on these issues, particularly determining the optimal scheduling and speeds of automated vehicles to cross intersections safely, without stopping and without the need for a traffic signal. To accomplish this, the problem is formulated as a mixed-integer linear programming (MILP) optimization problem for a generic intersection characterized by an arbitrary number of road segments and lanes. In addition, a discussion of the properties of the problem solutions, an application of the proposed approach to a case study, a solution strategy that can be used to solve large problem instances in a reasonable time, and a sensitivity analysis of the primary model parameters are provided. The considered case study shows that the proposed model can effectively avoid vehicle conflicts and increase the intersection capacity up to double with respect to both first-come first-served control policy and signalized intersections.

1. Introduction

It is now recognized worldwide that automated vehicles will play a key role in future road transport policies because they address many important societal transport challenges, such as road safety, congestion, accessibility and social inclusion. In addition, the key element to tackle such challenges is often claimed to be considering the road transport system as a whole: vehicles, infrastructure, and users. In such a holistic view, the introduction of automated driving should be considered from a fourfold perspective: transport policy issues; technological challenges; legal and regulatory frames; and digital transformation. In this connection, the continuous development of suitable technologies allows a gradual introduction of growing levels of automation in different sectors of road transport to make the system more efficient, safer, and environmentally friendly.

The SAE International J3016 ‘Levels of Driving Automation’ standard (SAE International, 2021) performed substantial work on standardization with the goal of providing a common terminology for automated driving, a harmonized classification system and supporting definitions. In particular, the J3016 standard identifies six levels of

driving automation from *no automation* to *full automation*, clearly addressing the levels corresponding to the so-called *automated vehicles* (i.e., levels 4—High Automation and 5—Full Automation). Additionally, it provides basic definitions and levels of vehicle functional aspects of technology, describes categorical distinctions for a stepwise progression through the levels that are consistent with current industry practice, and clarifies for each level what role (if any) drivers have in performing the dynamic driving task while a driving automation system is engaged.

In this framework, confined areas with restricted access control and dedicated lanes are more suitable to test and introduce *automated vehicles* with respect to having them deployed to open roads with mixed traffic flows. In such a dedicated context, the use of automation and communication technologies would even allow the external coordination of vehicles to avoid collisions and achieve the highest possible crossing performance. Such a scenario is made possible by the implementation of the so-called *smart roads*, concept that refers to the use of automation and communication technologies applied to the road infrastructure. Smart roads allow vehicles to exchange information not

* Corresponding author.

E-mail addresses: federico.gallo@unige.it (F. Gallo), alessandra.milagostovich@gmail.com (A. Miagostovich), davide.giglio@unige.it (D. Giglio), angela.difebbraro@unige.it (A. Di Febbraro), nicola.sacco@unige.it (N. Sacco).

<https://doi.org/10.1016/j.ejtl.2025.100158>

Received 23 February 2023; Received in revised form 8 April 2025; Accepted 24 April 2025

Available online 17 May 2025

2192-4376/© 2025 The Authors. Published by Elsevier B.V. on behalf of Association of European Operational Research Societies (EURO). This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

only with each other, but also with the infrastructure (by means of sensors, vehicular ad-hoc networks, and road side units), and have attracted growing attention thanks to their potential to increase efficiency (e.g., better use of infrastructures), safety (e.g., vehicle collision avoidance and speed control) and sustainability (e.g., reduction of congestion and of pollutant emissions) of the transport sector (Pompigna and Mauro, 2022). In particular, smart roads populated only by autonomous vehicles allow the exploitation of such technologies at the highest level, allowing the optimal control of traffic to reduce congestion, travel times and emissions (Shladover, 2021). Thus, to fully take advantage of this concept, there is an ever growing need of management and coordination methods for automated vehicles, especially at intersections, which are a critical element of road networks due to the limited space vehicles have to share. In this framework, optimization models can be particularly of interest, since they allow to efficiently deal with vehicle collisions and safety while minimizing some desired variables, such as travel times or delays. This study aims to exploit the opportunities mentioned above to ensure the best use of physical infrastructures and to provide evidence of the potentialities of the smart roads.

In addition, even if the proposed approach is applied in this study to road transport, it is sufficiently general to be applied to other scenarios that allow a spatial restriction of a portion of infrastructure to a specific type of vehicle, such as autonomous navigation in port areas, which is expected to become reality in the near future (Martelli et al., 2021).

This study addresses the circulation of automated vehicles in reserved roads, and focuses on the optimization and coordination of vehicle crossings at unsignalized intersections, following the common belief that when the traffic flow will only consist of fully automated vehicles, no traffic lights will be necessary. More in details, the main contribution of this paper is a model for the combined optimization of both vehicle schedules (i.e., passing orders) and motion planning (i.e., speeds over given trajectories) inside intersections. The model is formalized as a mixed-integer linear programming (MILP) model, which has the advantage of easily dealing with high-granularity space discretization. This, in turn, allows an optimal use of the space in the intersection and to address vehicle conflicts by determining all the pairs of incompatible positions, i.e., positions in the intersection that cannot be occupied simultaneously by the vehicles. The major drawback of MILP problems is that the computational time increases fast with the number of variables; for this reason, the paper proposes also a rolling horizon-based solution approach which is able to deal with high-demand scenarios and find near-optimal solutions in reasonable time.

The remainder of this paper is organized as follows. Section 2 provides a literature review and the primary contributions of this study. Section 3 introduces the considered problem, its architecture, and the modeling approaches adopted. Section 4 describes the mixed-integer linear programming formulation of the optimization model. Section 5 discusses the primary characteristics of the obtained solutions, the computational performance of the model and a comparison with a real intersection control policy. The paper ends with some concluding remarks (Section 6) and with an appendix with a detailed analytical formalization of the vehicle trajectories (Appendix).

2. Literature review and paper contributions

Intersections are a commonly studied topic in transportation engineering (Findley, 2015) because they cause congestion issues (since they tend to become bottlenecks of traffic flow), and safety issues; for example, approximately 40% of vehicle collisions in the United States occur where two or more roads merge (Choi, 2010). Thus, intersection management strategies, especially those considering the presence of connected and automated vehicles (CAVs), are critically important, since they promise to provide improvements in intersection performance while increasing safety. Therefore, this topic has received a lot of attention from the research community (Li et al., 2014a).

Earlier intersection management strategies focused on signalized intersections, where autonomous vehicles have to interact with a traffic light, and signal control algorithms are developed (Guler et al., 2014; Yang et al., 2016), also in combination with the optimization of vehicle paths (Li et al., 2014b). In addition, Feng et al. (2019) describe a method based on a green light optimal speed advisory (GLOSA) that minimizes vehicles' intersection arrival time during a green phase. However, if vehicles are connected and able to communicate with themselves and the infrastructure, a traditional traffic light is no more necessary, as vehicles can be coordinated by an intersection controller, which determines the driving strategy to follow. In this way, the potentialities of CAVs technologies can be fully exploited, and the capacity of intersections can significantly increase. Autonomous intersection management (AIM) addresses the problem of coordinating CAVs inside intersections, exploiting the automation and communication technologies of both vehicles and infrastructures. Nonsignalized AIM strategies have been studied in Kamal et al. (2013) and Kamal et al. (2015). These methods require a coordination scheme of automated vehicles at an intersection without using any traffic lights, and an intersection coordination unit uses two-way communication to receive basic driving information from the approaching vehicles (e.g., current position, speed, and destination) at the approaching intersection, and sends guidance to them after having determined their control inputs. Additionally, Chen et al. (2020) propose a conflict decision model, based on game theory, to allow intelligent vehicles to select the optimal driving strategies (in the sense of Nash equilibrium) to maximize the overall driving revenue based on safety, efficiency, and comfort.

AIM strategies can be further classified (Abbas-Turki et al., 2023) according to the adopted architecture and protocol (i.e., how conflicting CAVs share the common space). The architecture can be centralized or decentralized (Yang and Monterola, 2016; Rodrigues de Campos et al., 2017): while in decentralized architecture vehicles can self-organize their entrance in the intersection (or conflict zone), in centralized architecture an intersection manager communicates and manages all the vehicles, guaranteeing more safety (related to the collision avoidance) and intersection performance.

As for the protocols, the first one to be proposed is stop and go, where CAVs can only receive 'stop' or 'go' sign. This protocol can be considered an extension of ordinary traffic lights, and it can also be referred to as virtual traffic light (Bazzi et al., 2016). More advanced protocols include virtual platoon, where vehicles are grouped to cross together the common space, and reservation-based protocols, where each CAV receives the time instant at which it is allowed to enter the conflict zone, and follows it. Some studies proposed a combination of protocols, as in Sai  ns-V  zquez et al. (2018), where authors propose a combination of virtual traffic lights and platooning, where vehicles near the intersection can form platoons to reduce the crossing time.

Vehicle scheduling is then used to define the way CAVs cross the intersection; two families of approaches are present in the literature, the optimization-based ones, and the rule-based ones. The first ones, also known as exact and heuristic approaches, compute the optimal (or near-optimal) vehicle passing sequences and times to enter the shared space. The second ones, also known as policy-based approaches, define simple rules to determine the passing orders; these include first-come first-served (Dresner and Stone, 2004), which is the most popular one, first ready out (Zhang et al., 2015), time to react (Kloock et al., 2019), and distributed clearing policies (Wu et al., 2018).

2.1. Optimization-based AIM strategies

The optimization-based methods allow to overcome the limitations of rule-based methods, changing the way the space reservation in the intersection is assigned to vehicles. In particular, optimization-based methods consider all vehicles at the same time to minimize variables of interest such as travel time, delay, or fuel consumption.

Existing studies differ on the intersection modeling approach (i.e., on the adopted discretization, on its granularity, on the trajectory modeling and on the collision region modeling), on the optimized variables, on the considered minimization objectives and constraints, and on the adopted formulation (Chen and Englund, 2016). With reference to the collision region modeling, two main approaches exist: the tile-based, where the intersection is divided into cells, and each of them can usually be occupied by only one vehicle at the time, and the conflict point-based, where instead of dividing the intersection into cells, the possible conflicting points of the vehicle trajectories are considered. As for the optimized variables, an important distinction can be done depending on whether the model optimizes only the vehicle passing orders or entrance times (i.e., it addresses a scheduling problem), or if it optimizes the vehicle speed or acceleration inside the intersection on a given trajectory (thus solving a motion planning problem).

In Tallapragada and Cortés (2015), the problem of coordinating the passage of vehicles through an intersection is studied with the goal of minimizing the total travel time; the intersection manager communicates with vehicles heading toward the intersection, groups them into clusters, and determines an optimal order of passage and average speed profiles.

Many studies have directly aimed at reducing fuel consumption or energy to develop the optimization model. However, the majority of them are stated as optimal control problems (Malikopoulos et al., 2018; Chalaki and Malikopoulos, 2022; Xu et al., 2022), which are hard to solve and performed singly for each vehicle.

Other recent studies have focused on methods based on cellular automata models, reinforcement learning, and risk functions to minimize the crossing time of vehicles while preventing collisions. Cruz-Piris et al. (2019) describe an automatic intersection optimization system for modeling an intersection, calculating the shortest paths of vehicles with minimum conflict points and optimizing, with a genetic algorithm, the arrival patterns to maximize the traffic rate. Bichiou and Rakha (2019) propose a real-time optimization problem where conflicts are avoided by dividing the intersection into four zones, each of which is assigned to at most one vehicle per time. Liu et al. (2019) introduce a cooperative scheduling mechanism for assigning priorities to vehicles, planning their trajectories and finding an entering window that avoids collisions through the definition of conflict zones and reduces delays; the speed is computed by the vehicles, which can also arrange their trajectory. Guan et al. (2020) use reinforcement learning to consider a centralized control scheme while reducing computational time. Collision avoidance is addressed through the definition of a risk function to be minimized; however, it is not guaranteed that conflicts are totally avoided. Liu et al. (2020) propose a coordination scheme with a multicollision-set strategy to allow one vehicle to simultaneously enter the intersection, thus avoiding conflicts. Karthikeyan et al. (2022) propose an intersection management system based on deep reinforcement learning to schedule vehicle crossings, while Gu et al. (2023) solve a similar problem by means of timed Petri nets. Finally, Cardei et al. (2024) propose a fastest trajectory planner for vehicle path and motion planning at intersections.

2.2. MILP-based AIM strategies

A class of optimization-based AIM strategies are those based on the formalization of a linear mathematical programming problem, usually using mixed-integer variables (MILP problems), as done in this study.

As it emerges from the literature reviewed in this section, MILP is a popular way to address AIM. Similarly to the optimization-based methods in general, MILP allows to achieve better performance than rule-based methods, where conflicts are resolved given a predefined rule (Wu and Qu, 2022). MILP requires usually a discretization of space or time; for this reason, it is particularly suitable to address vehicle conflicts, since after space discretization (e.g., by dividing the intersection plane into cells, or by discretizing the vehicle trajectories

into sequences of points, as done in this work), constraints can be easily written to prevent vehicles from occupying forbidden positions. MILP problems are easier to address with respect to nonlinear problems. For this reason, some nonlinear formulations are turned into linear (Zhu and Ukkusuri, 2015). In addition, they can be easily implemented and solved by different commercial solvers. The major drawback of MILP is that the computational time increases exponentially with the number of vehicles and lanes (Abbas-Turki et al., 2023), so usually only small instances of the problem can be solved by the commercial solvers. In all the other cases, an alternative solution approach must be developed. Finally, since they provide a discrete solution (e.g., in this work, times at which a vehicle must cross a point of its trajectory) there can be the necessity to make the solution continuous (e.g., by interpolation).

Table 1 shows a summary of the works that are most closely related to the one presented in this paper, i.e., MILP-based AIM strategies formalized as vehicle scheduling problems: the scheduling and motion planning main features in the intersection are highlighted, together with the modeling approach (i.e., modeling of conflict area, analytic trajectory, and granularity of discretization). With reference to the granularity of discretization, three levels are considered, ranging from low (referring to the entire intersection considered as a unique conflict area, or only trajectory conflict points considered) to high (space discretization lower than 1 m). In addition, focus is also on highlighting whether vehicle dynamics (i.e., mass, and adherence), kinematics constraints (i.e., specific constraints on acceleration and speed), and exact vehicle size are considered.

In Zhu and Ukkusuri (2015), a linear programming formulation for autonomous intersection control that considers traffic dynamics within a connected vehicle environment is proposed. However, it optimizes only macroscopic variables (i.e., vehicle flows). In Müller et al. (2016), the obtained vehicle schedules are then used as an input for a motion planning problem, solved analytically with linear motion equations, to allow vehicles enter the intersection at the designated time. Within the intersection, vehicles are then assumed to travel at constant speed. Fayazi and Vahidi (2018) determine the optimal entering times at the intersection to minimize delays. However, turnings are not allowed, and a constant speed limit inside the entire intersection is considered. Ge et al. (2021) address the problem of centralized coordination by maximizing the vehicle speeds while grouping vehicles into clusters to reduce computational times.

Levin and Rey (2017) determine the entrance time of vehicles at intersection, minimizing the total exit time. Inside the intersection, vehicles are assumed to travel at constant speed. Both Wu et al. (2022) and Yu et al. (2019) focus only on the vehicle scheduling to minimize vehicle delay and find the vehicle entering times (at the intersection) and passing orders. In addition, Wu et al. (2022) determine also the entering and exiting lanes for each vehicle. Both Ma and Li (2021), Yao et al. (2022) and Yao et al. (2023) address, in addition to the scheduling problem, also the speed trajectory optimization (i.e., determine acceleration and speed). However, Ma and Li (2021), Yao et al. (2023) address the problem only before vehicles enter the intersection, while Yao et al. (2022) propose a control theory-based method, to minimize fuel consumption, solved for each vehicle. Liu et al. (2023) optimize vehicle entrance times at the intersection. Vehicles can stop before entering the intersection, but then they travel at a constant speed selected among a predefined set. In addition, trajectories minimizing the likelihood of conflicts are selected.

2.3. Literature gaps and paper main contributions

From the analyzed literature, the following limitations emerge. The vehicle scheduling and motion planning problems have been mainly addressed as separate problems; within the AIM framework, the majority of the studies consider only the vehicle scheduling (Fayazi and Vahidi, 2018; Cruz-Piris et al., 2019; Zhu and Ukkusuri, 2015; Wu et al., 2022; Yu et al., 2019). In the works that addressed both the scheduling and

Table 1
Summary of most closely related works.

<i>Study</i>	<i>Scheduling goals</i>	<i>Motion planning</i>	<i>Objective function</i>	<i>Conflict avoidance</i>	<i>Analytic form of trajectories</i>	<i>Granularity of discretization</i>	<i>Vehicle dynamics</i>	<i>Kinematics constraints</i>	<i>Vehicle size</i>	<i>Solution approach</i>
Zhu and Ukkusuri (2015)	Flows per time unit	–	Travel time	Conflict points	–	Low	–	–	–	–
Müller et al. (2016)	Entrance times	Only before intersection	Delay	Conflict area	–	Low	–	–	✓	CBC solver
Levin and Rey (2017)	Entrance times and sequences	Yes, but vehicles travel at same speed	Exit time	Conflict points	✓	Low	–	–	Only length	Rolling horizon
Fayazi and Vahidi (2018)	Entrance times and sequences	–	Delay	Conflict area	–	Low	–	–	–	CPLEX optimization package
Yu et al. (2019)	Entrance times and passing orders	–	Delay	Conflict tiles (cells)	✓	Medium	–	–	✓	Gurobi solver
Ge et al. (2021)	Passing orders at conflict points	Yes (speed over predefined trajectories)	Speed ^a	Conflict points	–	Low	–	–	– ^b	Ad-hoc algorithm
Ma and Li (2021)	Entrance times	Only before intersection	Max travel time of all vehicles	Conflict tiles (cells)	–	Medium	–	✓ ^c	– ^b	CPLEX solver
Yao et al. (2022)	Entrance time at conflict zones	Yes, solved for each vehicle	Delay	Conflict area	–	Low	✓	✓	Only length	CPLEX + nonlinear programming and infinitesimal method
Wu et al. (2022)	Entrance times, and entering and exiting lanes	–	Delay	Conflict tiles (cells)	–	Medium	–	–	✓	Ad-hoc heuristic method
Liu et al. (2023)	Entrance times	Yes, but vehicles travel at same speed	Delay	Conflict tiles (cells)	–	Medium	–	–	✓	CPLEX solver
Yao et al. (2023)	Entrance times	Only before the intersection	Delay	Conflict regions	–	Medium-High	–	✓	✓	Ad-hoc heuristic
Hadjigeorgiou and Timotheou (2024)	Entrance times	Only before the intersection	Travel time	Conflict area	–	Low	–	✓ ^c	Only length	Gurobi solver
Proposed model	Entrance time and passing orders at each conflict	Yes (in the entire intersection)	Travel time and speed variations	Incompatible points	✓	High	✓	✓	✓	Rolling horizon-based ad-hoc method

^a Maximized within specified bounds.

^b Not explicitly.

^c Only fixed bounds on acceleration and speed.

motion planning, the latter is usually solved in sequence, i.e., with the vehicle schedules as an input (Yao et al., 2022, 2023), in a decentralized way, individually for each vehicle (Liu et al., 2019; Bichiou and Rakha, 2019; Yao et al., 2022; Tallapragada and Cortés, 2015), only in the road stretch before the intersection (Müller et al., 2016; Ma and Li, 2021; Hadjigeorgiou and Timotheou, 2024), or assuming constant speed inside the intersection (Levin and Rey, 2017; Müller et al., 2016; Liu et al., 2023; Yao et al., 2023).

Additionally, few papers consider vehicles with their exact spatial dimension (especially in conflict point-based methods and in tile-based low-granularity discretization approaches). In fact, conflict point-based methods usually state the problem as one-dimensional, without considering the entire vehicle size. For example, Levin and Rey (2017) propose a conflict point-based approach that considers only points where trajectories intersect each other. On the other hand, tile-based approaches (which divide the intersection into cells) are difficult to integrate in a MILP formulation, and high-granularity of the intersection discretization can worsen the computational tractability of the problem. For this reason, most of the studies consider low or medium-granularity, leading to a nonoptimal space management in the intersection.

Besides, few studies have considered real form of the trajectories and kinematics constraints (e.g., constraints to limit acceleration and speed). For instance, adequate speed limits, in particular as a function of the curvature of the trajectory and the pavement-tire friction ratio, can guarantee the stability of the vehicle and a safe cross of the intersection in any condition, whereas acceleration bounds can be used to guarantee enough comfort for the users, as well as a reduction in the energy consumption. Vehicle dynamics (i.e., mass, adherence, and stability) have been mainly considered only in optimal control problems, which are difficult and solved individually for each vehicle.

To the best of the authors' knowledge, no studies have considered in an integrated way and within a unique problem all the aspects mentioned above. Lastly, analyzed studies adopted a variety of intersection layouts, but the majority of them consider the presence of dedicated lanes (i.e., lanes from which only specific maneuvers at the intersections can be done), so do not fully exploit the potentialities of CAVs and AIM, where, in theory, all maneuvers could be done from each lane.

This paper is aimed at filling the above mentioned gaps, and its primary contributions are as follows.

First, the optimization problem is formalized as an MILP problem whose solution generates the optimal speeds along the assigned trajectories that are chosen among a predefined set according to the destinations for all the vehicles crossing the intersection, minimizing the total travel time while avoiding vehicle stops near or inside the intersection. Thus, the problem is formulated for a generic intersection layout characterized by an arbitrary number of roads and lanes, while the trajectories are fully determined analytically. As will be clarified, by only optimizing vehicle speeds, the sequence of vehicles crossing the intersection also results in the best schedule; consequently, both vehicle scheduling and motion planning are solved within the same problem. The key features of the MILP formalization are summarized as follows: vehicles are considered with their real space occupancies, positions, speeds and accelerations; vehicle trajectories are modeled to consider their geometric characteristics such as the curvature radius and the consequent speed limits; kinematics constraints are used to specify speed and acceleration bounds. The speed limits are derived as a function of the local curvature of the trajectory, and take into account also the vehicle mass and the friction coefficient between the pavement and the tires. In this way, different coefficient values could be used depending on the pavement type and weather conditions (e.g., to ensure additional safety during a snowy or rainy event). The acceleration bounds are used to guarantee enough comfort for the users, and can also reduce the energy consumption. Conflicts are avoided by specific constraints that manage the incompatibilities of some relative positions of vehicles

inside the intersection. The modeling approach and the incompatibility constraints allow to deal with high-granularity discretizations, which allow higher detail modeling and better space use, and overcome the limitations of conflict points and tile-based approaches. In addition, the proposed model allows to make all maneuvers (turnings) from each lane, without the need to specify dedicated lanes, thus fully exploiting the available space and possibilities of AIM and CAVs.

Second, this paper proposes an approach based on a rolling horizon framework to solve large instances of the problem in an acceptable amount of time.

Third, the paper presents simulation results for the performance comparison between the proposed method, a first-come first-served control policy, and a traditional traffic light control with optimized phase sequences and lengths.

This paper is part of a research activity on the management and optimization of automated vehicles inside road networks. Within this line of research, in a previous paper (Di Febbraro et al., 2020), the authors introduced the general idea of the proposed traffic management system for road networks, which are assumed to have a modular structure and to be composed of intersection elements and road elements connected together. Each element of the network is coordinated by a proper controller that determines the motion of the automated vehicles in it. That paper then describes the optimization model of the road element.

In contrast, this paper presents in detail the optimization model of the intersection element and discusses the analytical formulation of the vehicle trajectories and the addressed methodology to find the solution. The latter can be divided into three steps: the spatial discretization of the trajectories and of the time; the MILP formulation and solution, which provides discrete-time speed values; and the continuous approximation of the solution, which allows to find the continuous-time speed functions to be provided to the vehicles.

In addition, the optimization model of the intersection usually requires a higher computational time to be solved than the optimization model of the road element due to the higher number of variables and vehicle flows. Thus, this paper describes in detail an algorithm to find a solution in an amount of time comparable to the addressed application and discusses the relationships between computational performances and result performances. These results are also compared with the results obtained with a traditional traffic light-based approach to provide evidence of the gain in terms of performance that can be achieved by the proposed approach.

Finally, an analysis of the relevance of the primary model parameters on the performance of the solution and on the computational times is provided.

3. Problem definition and methodology

This section first introduces the considered problem, the proposed system architecture, and the followed methodological approach, using the notation reported in Table 2. The main goal is to propose an AIM system that determines the optimal passing orders, trajectories, and speeds of connected and automated vehicles to allow them to safely cross an intersection, avoiding conflicts, and ensuring vehicle stability and comfort for the passengers. In doing so, the aim is to minimize the total travel time of the vehicles inside the intersection, together with excessive speed variations.

To this aim, some general definitions are given. Thus, let consider a generic intersection (as the one depicted in Fig. 1), and let:

- γ_i be the locus of admissible successive positions on the intersection plane of the barycenter of vehicles in the same stream i , which, for the sake of compactness, is hereafter referred to as *trajectory*. Each point $P \in \gamma_i$ is in one-to-one correspondence with its distance $\ell(P)$ from the initial point of γ_i ($P \Leftrightarrow \ell(P)$, see Appendix);

Table 2

Notation.

Symbol	Meaning
S	set of streams crossing the intersection
$\mathcal{V}_i$	set of vehicles on the stream i . Such a set is sorted; thus, the direct follower of the vehicle v is $(v+1)$
$\gamma_i, \forall i \in S$	trajectory followed by vehicles in the stream i (see Appendix)
$\Gamma = \{\gamma_i \forall i \in S\}$	set of all trajectories
$P_{i,0}$	position of the trigger related to γ_i
$P_{i,1}$	first point of γ_i considered in the optimization problem
$P_{i,h}$	generic point of $\gamma_i, i \in S$
$P_i = \{P_{i,h}\}$	set of n_i points along γ_i
$P_{i,h} \leftrightarrow P_{j,k}$	the points $P_{i,h}$ and $P_{j,k}$ are incompatible
$a_{i,h}$	generic arc of the trajectory γ_i between two consecutive points $P_{i,h}$ and $P_{i,h+1}$
δ	fixed length of the arc $a_{i,h}$
$\ell_{i,h}$	distance of $P_{i,h}$ from $P_{i,1}$
$u_{rd}^{\min}$ and $u_{rd}^{\max}$	minimum and maximum admissible vehicle speed in the road stretches between the triggers and the intersection
$u_{int}^{\max}$	maximum admissible vehicle speed inside the intersection
$\Phi_{\min} = \delta / u_{int}^{\max}$	minimum travel time on an arc
Φ_l and Φ_h	maximum allowed relative vehicle travel time variations when the speed is low and high, respectively
$r_{i,h}$	coefficient determined off-line that reduces the maximum allowed speed on the arc $a_{i,h}$ (see Section 3.4)
α	weighting coefficient in the objective function of the MILP problem
β	safety parameter used to increase the minimum permitted distance between vehicles (additional headway)
$\tau \in \mathbb{R}$	generic time instant
$s_{i,u,0}$	time instant at which vehicle u crosses the trigger and enters the road stretch between the trigger and the intersection
$s_{i,v,h}$	real variable indicating the time instant at which the vehicle v enters the arc $a_{i,h}$ of the trajectory γ_i
$t_{i,v,h}$	real variable indicating the time instant at which the vehicle v exits the arc $a_{i,h}$ of the trajectory γ_i
$x_{i,v,j,u}$	Boolean variable set to 1 if the vehicle v in the stream i has right of way with respect to the vehicle u in the stream j
$s_{i,v,h}^*, t_{i,v,h}^*, x_{i,v,j,u}^*$	optimal values of the variables $s_{i,v,h}, t_{i,v,h}, x_{i,v,j,u}$

- $\mathcal{N} = \{(i,j) | \gamma_i \cap \gamma_j = \emptyset, i \neq j\}$ be the set of indices of *nonconflicting trajectory pairs* (i.e., the trajectories that do not cross each other; e.g., γ_3 and γ_{18} in [Fig. 1](#));
- $\mathcal{C} = \{(i,j) | (i,j) \notin \mathcal{N}, P_{i,1} \neq P_{j,1}, i \neq j\}$ be the set of indices of *conflicting trajectory pairs* (i.e., the trajectories that cross each other only at some points; e.g., γ_3 and γ_6 in [Fig. 1](#));
- $\mathcal{O} = \{(i,j) | (i,j) \notin \mathcal{N}, P_{i,1} = P_{j,1}, i \neq j\}$ be the set of indices of *partially overlapped trajectory pairs* (i.e., the trajectories that initially overlap; e.g., γ_2 and γ_3 in [Fig. 1](#)).

3.1. Assumptions

The following assumptions hold:

1. The considered intersection is characterized by a set of geometrical trajectories $\gamma_i, i \in S$, followed by streams of vehicles. Such trajectories, which depend only on the intersection layout, lay in the intersection plane and are analytically predetermined as described in [Appendix](#). An example is shown in [Fig. 1](#), which shows a generic 4-way 2-lane intersection and the predefined trajectories associated with the 24 possible streams, but other layouts could be considered. Each trajectory is in a one-to-one relation with a vehicle stream. It is worth pointing out that, given the above considerations, more than one vehicle can be assigned to the same trajectory occupying, at the same time, only compatible positions;
2. Vehicles are not allowed to change stream or to overtake the others in the same stream; therefore, vehicles are sorted in each stream;
3. Vehicles are considered with their real space occupancy, and it is possible to identify off-line the set of reciprocal incompatible positions, as described in [Section 3.5](#);
4. Vehicles have an SAE level 4 or 5 of automation and thus can safely cross an intersection with the optimal speeds and accelerations determined by solving the proposed MILP problem without the need for any human interaction;
5. Vehicles can communicate with the infrastructure through specific on-board vehicle-to-infrastructure (V2I) devices to provide information about their travel to the intersection controller and to receive from it the optimal speeds. Safety and comfort in crossing the intersection are ensured by specific constraints of the optimization problem;
6. When a vehicle u crosses a trigger located at a distance Δ_i from the intersection, it sends the crossing time $s_{i,u,0}$ to the intersection controller. A sketch of the trigger is shown in [Fig. 2](#);
7. The speed limits along the incoming road stretch guarantee that all the vehicles can arrange their speed to reach the first point of the trajectory γ_i at the time instant determined by the optimization problem.

3.2. System architecture

A centralized control architecture is used in which an intersection controller solves, at fixed discrete-time instants, a single optimization problem that considers all vehicles that will reach the intersection in the next interval. To be more precise, the optimization problem that is solved at the generic discrete time instant $\tau_k - \epsilon$ considers all the vehicles whose Estimated Arrival Time at Intersection (EATI) is in the interval $[\tau_k, \tau_{k+1}) = [\tau_k, \tau_k + \Delta\tau)$, with $\Delta\tau$ being the problem time window and ϵ the maximum time allowed to the intersection controller for performing the steps reported below and providing the speeds to the vehicles.

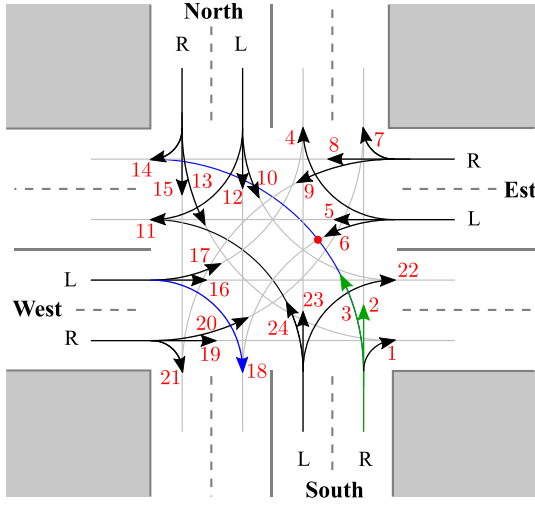


Fig. 1. General sketch of an intersection with 24 admissible streams indicated by arrows. The gray dashed lines represent the related trajectories, the blue trajectories exemplify two nonconflicting pairs $(\{3, 18\} \in \mathcal{N})$, the red dot identifies an example of conflict $(\{3, 6\} \in \mathcal{C})$, and the green dots exemplify two partially overlapped trajectories $(\{2, 3\} \in \mathcal{O})$.

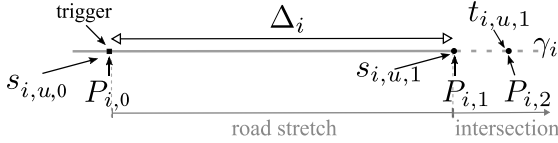


Fig. 2. Trigger located at a distance Δ_i from the beginning of trajectory γ_i in $P_{i,1}$.

The logical steps that are accomplished by the system in the interval $[\tau_k - \epsilon, \tau_{k+1})$ are as follows:

1. All vehicles that are no more than a suitable distance d from the intersection are queried by the intersection controller about their EATI, origin, destination, and shape (i.e., width, length and barycenter position);
2. The intersection controller defines the set $\mathcal{V}_k$ of the active vehicles gathering all vehicles with $\text{EATI} < \tau_{k+1}$ and, given their origins and destinations, identifies the stream of each vehicle among those allowed in the intersection (e.g., with reference to the intersection sketched in Fig. 1, one out of the admissible 24);
3. The intersection controller states the optimization problem, solves it and communicates the solution to each vehicle in the set $\mathcal{V}_k$;
4. The vehicles in set $\mathcal{V}_k$ safely cross the intersection by traveling at the optimal speed along the curve associated with the stream identified at step 2 with the speeds determined at step 3.

Such an optimization scheme is applied repetitively interval by interval every $\Delta\tau$ seconds. The speeds received by vehicles (at τ_k) that are still inside the intersection at τ_{k+1} represent constraints for the problem instance referred to the interval $[\tau_{k+1}, \tau_{k+2})$, as discussed in Section 4.2.

The parameter d can be set to $2\hat{v}\Delta\tau$, where $\hat{v}$ is the average speed of vehicles arriving at the intersection; thus, d is the space traveled on average in $2\Delta\tau$ by vehicles. The parameters ϵ and $\Delta\tau$ can be suitably set in accordance with computational requirements, as discussed in the case study described in Section 5. However, ϵ must be lower than $\Delta_i/v_{\text{rd}}^{\text{max}}$; thus, vehicles must receive the speeds from the intersection controller before entering the intersection. If this condition is not satisfied, a backup strategy must be used and can consist of using the best admissible solution found by the intersection controller within ϵ . If

neither an admissible solution is found, to preserve safety, only vehicles traveling along nonconflicting trajectories can be allowed to cross the intersection during $[\tau_k, \tau_k + \Delta\tau)$.

3.3. Spatial discretization

The considered optimization problem aims to minimize the total travel time of a given set of vehicles in a given time interval while guaranteeing vehicle safety. Such a problem is intrinsically a continuous-time optimization problem because the state of each vehicle v is represented by differential state equations providing its position and speed for any time instant τ . In this paper, such a problem is reformulated as a mixed-integer linear programming (MILP) problem, where trajectories have been discretized into *sequences of arcs* of the same length δ . The search for a suitable discretization is an off-line problem to solve once per intersection, and the choice of $\delta = 0.5$ m is sufficient for the intersection considered in this paper.

Due to the spatial discretization, it is possible to introduce the sorted sequences $P_i = \{P_{i,h}, h = 1, \dots, n_i\}$, $\forall i \in \mathcal{S}$ of points of the trajectory γ_i such that $\ell_{i,h+1} > \ell_{i,h}$ and $\ell_{i,h+1} - \ell_{i,h} = \delta$, where $\ell_{i,h} = \ell(P)|_{P=P_{i,h}} = \delta \cdot (h-1)$ and $\ell(P)$ is the function distance of P from $P_{i,1}$ (see Appendix). Based on this definition, $\ell_{i,h} \Leftrightarrow P_{i,h}$, and n_i depends on the total length of trajectories and on the discretization parameter δ . For simplicity trajectories are assumed to be adjusted so that $\ell_{i,n_i} \bmod \delta = 0$, $\forall \gamma_i \in \Gamma$.

3.4. Curved trajectories and speed limits

In this section, the maximum admissible speed of vehicles in any point of an arc is discussed with reference to its dependance on the road characteristics and, in particular, on the geometry of the trajectories. In fact, given the maximum admissible speed along a straight trajectory inside the intersection, $v_{\text{int}}^{\text{max}}$, a ‘local’ reduction value must be determined for each point $P_{i,h} \in \gamma_i$ as a function of the trajectory curvature in $P_{i,h}$. To formalize such a requirement, a speed reduction factor $r(P)$ is first determined as a function of the point P , as described in Appendix. Then, a specific value is associated with each arc $a_{i,h}$ by choosing the minimum along the arc: $r_{i,h} = \min_{P \in a_{i,h}} r(P)$. All the arcs corresponding to straight trajectory stretches have $r_{i,h} = 1$ (i.e., no speed reduction is required).

The functions $r(P)$ can be determined for any possible trajectory of an intersection. In this paper, they have been determined for the four trajectories in Fig. 3, which are congruent to the other curved trajectories in the intersection in Fig. 1 (i.e., all the curved trajectories can be obtained by those in Fig. 3 via rotation and translation). All the trajectories in Fig. 3 are made up of:

1. A line from $P_{i,1}$ to A_i (infinite curvature radius);
2. A clothoid from A_i to B_i (variable curvature radius from infinite to a finite, nonnull, value);
3. A circular segment from B_i to D_i (finite, nonnull, curvature radius);
4. A clothoid from D_i to E_i (variable curvature radius from a finite, nonnull, value to infinite);
5. A line from E_i to P_{i,n_i} (infinite curvature radius).

Therefore, $r(P)$ is equal to 1 (no speed reduction) for the initial and final parts, decreases from 1 to a minimum value in the clothoids, and has a constant minimum value for the circular segments because the higher the curvature, the lower the value of $r(P)$.

A sketch of the reduction factors for the curves in Fig. 3 is shown in Fig. 4 where for clarity, the points A_i , B_i , C_i , D_i , and E_i are indicated.

The considerations described in this section are general because any trajectory along a real world road can be described as a sequence of lines, clothoids, and circular segments.

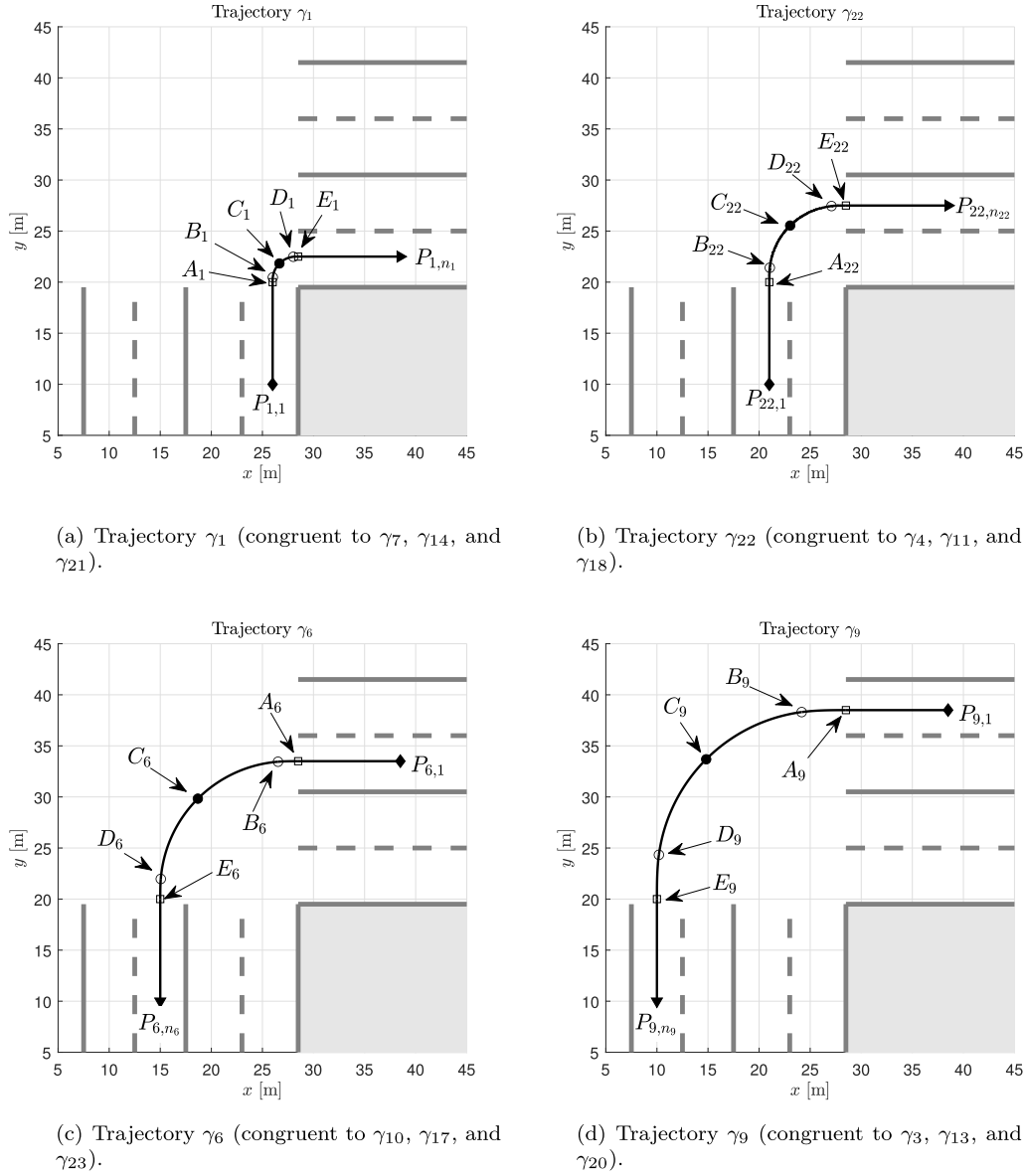


Fig. 3. Representation of the four types of curve trajectories in the intersection shown in Fig. 1. The points A_i , B_i , C_i , D_i , and E_i correspond to those in Fig. 4.

3.5. Incompatibility constraints and conflict management

Vehicle collision avoidance is ensured by MILP constraints that prevent vehicles from reaching two incompatible points concurrently. This section describes in detail the concept of ‘incompatibility’ and discusses how pairs of incompatible nodes are identified and used in the MILP.

First, the space occupancy of each vehicle at any point $P_{i,h} \in P_i$, $\forall i \in S$, is defined as the space identified by a rectangular shape whose longer side is parallel to the unit tangent vector $T_{i,h}$ in $P_{i,h}$, assuming that the barycenter of the vehicle is at this point. An example of these occupancies is shown in Fig. 5 where the empty rectangles indicate the sequence of the consecutive occupancies of a vehicle traveling along γ_i , having assumed the previously discussed discretization. Each point P belonging to the arc $a_{i,h}$ is associated with a shape coherently positioned among those associated with the points $P_{i,h}$ and $P_{i,h+1}$.

Then, two generic points P and Q are incompatible (i.e., there cannot be two vehicles on P and Q concurrently), or $P \nleftrightarrow Q$, if the relevant occupancies overlap.

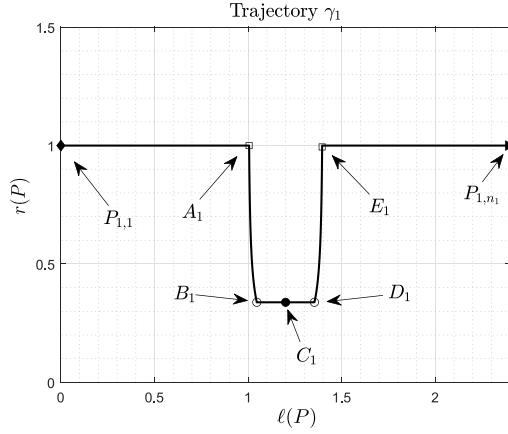
The above definition is extended to the points $P_{i,h} \in P_i$, $\forall i \in S$ in the following way: two points $P_{i,h}$ and $P_{j,k}$ are incompatible, or $P_{i,h} \nleftrightarrow P_{j,k}$,

if $\exists P \in a_{i,h}, Q \in a_{j,k} | P \nleftrightarrow Q$. This definition is conservative because, given a generic point $P \in a_{i,h}$ that is incompatible with a point Q , it considers incompatible with Q all the points of the arc $a_{i,h}$ before and after P . This definition also allows to write safety constraints only with references to the initial points $P_{i,h}$ of arcs and is not too restrictive thanks to the small value of δ .

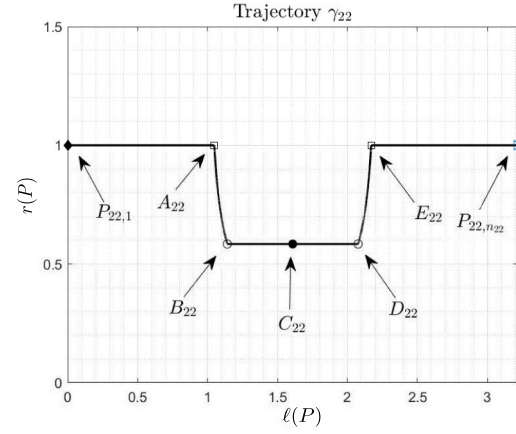
Given the above definitions, it is possible to distinguish between two cases: points belonging to the same trajectory and points belonging to two different trajectories.

3.5.1. Incompatibility of points belonging to the same trajectory

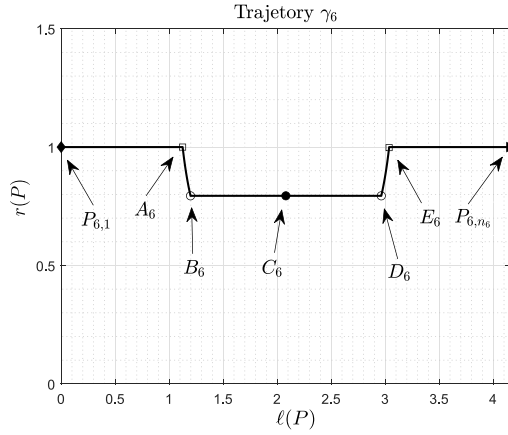
Every pair of incompatible points $P_{i,h} \nleftrightarrow P_{i,k}$ belonging to the same trajectory γ_i must be associated with a constraint that guarantees the safe spacing between a leader vehicle v and a follower vehicle $(v+1)$. An example is shown in Fig. 6, where the occupancies of vehicles v and $(v+1)$ are shown with colored rectangles positioned in $P_{i,k}$ and $P_{i,h-1}$, respectively, and the positions of $(v+1)$ in $P_{i,h}$ and $P_{i,h+1}$ are represented with continuous and dashed rectangles, respectively. In such a case, $P_{i,h}$ and $P_{i,k}$ are incompatible because, by traveling along arc $a_{i,h}$, vehicle $v+1$ may collide with vehicle v , which is traveling on arc $a_{i,k}$. Therefore,



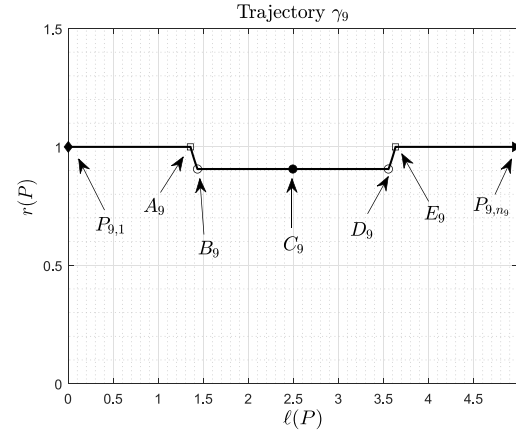
(a) Speed reduction factor along γ_1 (congruent to γ_7 , γ_{14} , and γ_{21}).



(b) Speed reduction factor along γ_{22} (congruent to γ_4 , γ_{11} , and γ_{18}).



(c) Speed reduction factor along γ_6 (congruent to γ_{10} , γ_{17} , and γ_{23}).



(d) Speed reduction factor along γ_9 (congruent to γ_3 , γ_{13} , and γ_{20}).

Fig. 4. Speed reduction factors of the trajectories shown in Fig. 3: the speed is reduced more and more as the curvature of the trajectories increases. The points A_i , B_i , C_i , D_i , and E_i correspond to those in Fig. 3.

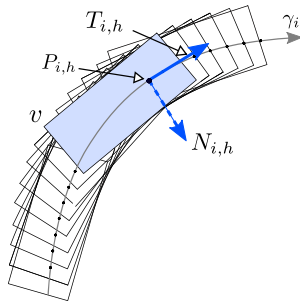


Fig. 5. Generic vehicle v traveling along a curved trajectory γ_i . The continuous and dashed arrows represent the tangent vector $T_{i,h}$ and the normal vector $N_{i,h}$ in $P_{i,h} \in \gamma_i$, respectively.

because vehicles cannot stop, the follower cannot even reach the point $P_{i,h}$ and must be slowed down before reaching it.

3.5.2. Incompatibility of points belonging to different trajectories

Every pair of incompatible points $P_{i,h} \leftrightarrow P_{j,k}$ belonging to the different trajectories γ_i and γ_j must be associated with a set of constraints that guarantees the assignment of the right of way to one and only one

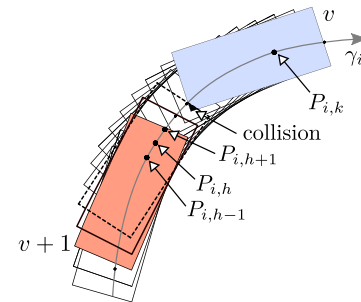


Fig. 6. Example of incompatibility between two points of the same trajectory $P_{i,h}$ and $P_{i,k}$: the follower vehicle ($v+1$) can reach $P_{i,h-1}$ but not $P_{i,h}$, while the leader vehicle v is on the arc $a_{i,k}$; otherwise, a collision may occur. Therefore, ($v+1$) must be delayed until v enters arc $a_{i,k+1}$.

of the vehicles in each pair of vehicles v and u traveling along γ_i and γ_j , respectively. An example of this second type of conflict is shown in Fig. 7, where the colored shapes indicate the positions of the vehicles on $P_{j,k}$ and $P_{i,h}$. In this specific frame, vehicle v obtained right of way

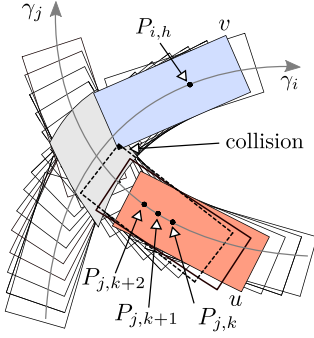


Fig. 7. Example of incompatibility between points of two conflicting trajectories. The colored shapes indicate the positions of vehicles u and v on $P_{j,k}$ and $P_{i,h}$; because $P_{j,k+1}$ and $P_{i,h}$ are not compatible, u can reach point $P_{j,k}$ and travel along arc $a_{j,k}$ but not along arc $a_{j,k+1}$.

and is on arc $a_{i,h}$; vehicle u can travel along arc $a_{j,k}$ but not along $a_{j,k+1}$, while v is on arc $a_{i,h}$; therefore, u must be slowed down before reaching point $P_{j,k+1}$.

3.5.3. Incompatibility sets

To guarantee the safe crossing of vehicles, a key element of the proposed approach consists of the identification of the sets of incompatible points:

$$\mathcal{I}^S = \{\{P_{i,h}, P_{i,k}\} | P_{i,h} \leftrightarrow P_{i,k}, P_{i,h}, P_{i,k} \in \mathcal{P}_i, i \in \mathcal{S}\} \quad (1)$$

$$\mathcal{I}^C = \{\{P_{i,h}, P_{j,k}\} | P_{i,h} \leftrightarrow P_{j,k}, P_{i,h} \in \mathcal{P}_i, P_{j,k} \in \mathcal{P}_j, (i, j) \in \mathcal{C}, j > i\} \quad (2)$$

$$\mathcal{I}^O = \{\{P_{i,h}, P_{j,k}\} | P_{i,h} \leftrightarrow P_{j,k}, P_{i,h} \in \mathcal{P}_i, P_{j,k} \in \mathcal{P}_j, (i, j) \in \mathcal{O}, j > i\} \quad (3)$$

which gather the pairs on incompatible points in the same stream, in conflicting streams (e.g., streams 3 and 6 in Fig. 1), and in partially overlapped streams (e.g., streams 2 and 3 in Fig. 1), respectively. The condition $j > i$ avoids duplicating the incompatibility pairs.

Thus, the incompatibilities of trajectories are analyzed off-line once to generate the sets $\mathcal{I}^S$, $\mathcal{I}^C$, and $\mathcal{I}^O$. A simple algorithm for determining these sets consists of the comparison of all the occupancy rectangles and has a computational complexity proportional to $|\mathcal{S}| \cdot (\max_{i \in \mathcal{S}} \{n_i\})^2$, which is proportional to the squared total number of points of the longest trajectory. These definitions can be used to determine the incompatibility sets for any possible combination of vehicle classes by considering rectangles of the proper sizes so that different vehicle types can be considered in the intersection.

4. The scheduling and motion planning model

In this section, the problem formulation is provided with the proposed heuristic solution approach and a discussion about the analytical characteristics of the optimal solution.

The considered problem is formulated using the real variables $s_{i,v,h}$ and $t_{i,v,h}$, which represent the instants at which the generic vehicle v begins and ends its travel along the arc $a_{i,h}$. Because all arcs have the same length δ , these two variables are related to the speed of the vehicle along the arcs, which is equal to $\delta/(t_{i,v,h} - s_{i,v,h})$. In addition, the Boolean variables $x_{i,v,j,u}$ allow to manage the incompatibilities defined in Section 3.5 and schedule whether vehicle v in stream i crosses the intersection before vehicle u in stream j ($x_{i,v,j,u} = 1$) or after ($x_{i,v,j,u} = 0$).

4.1. MILP formulation

Given the considerations and the modeling issues addressed in the previous section, the optimization problem can be formalized as follows:

$$[s^*, t^*, x^*] = \arg \min_{s, t, x} (\Theta + \alpha SF) \quad (4)$$

with:

$$\Theta = \sum_{i \in \mathcal{S}} \sum_{v \in \mathcal{V}_i} (t_{i,v,n_i} - s_{i,v,0}) \quad (5)$$

the total travel time. Since this definition takes into account the vehicle travel times both in the intersection and in the road stretch before it, possible vehicle queues on the approach to the intersection can be considered and minimized.

The second term of the objective function

$$SF = \sum_{i \in \mathcal{S}} \sum_{v \in \mathcal{V}_i} \sum_{h=1}^{n_i-3} |s_{i,v,h+3} - 2s_{i,v,h+2} + s_{i,v,h+1}| \quad (6)$$

represents a smoothing filter aiming at obtaining smooth speeds, avoiding sudden changes in acceleration that otherwise could appear in the solution discussed in Section 5.4 in the form of speed peaks of the type *high-low-high* or *low-high-low* as determined by *positive-negative-positive* or *negative-positive-negative* accelerations in a generic sequence of three arcs.

The meaning of the absolute value term in the smoothing filter can be explained as follows. Let consider a sequence of three generic arcs $a_{i,h}$, $a_{i,h+1}$, and $a_{i,h+2}$ along the trajectory γ_i . If $(s_{i,v,h+3} - s_{i,v,h+2}) < (s_{i,v,h+2} - s_{i,v,h+1})$, then the average speed along the arcs $a_{i,h+1}$ and $a_{i,h+2}$ is increasing. Analogously, considering the entire sequence of three arcs, if $(s_{i,v,h+3} - s_{i,v,h+2}) < (s_{i,v,h+1} - s_{i,v,h})$, then, on average, the vehicle is accelerating. Let:

$$\Delta\tau_{h,3} = (s_{i,v,h+1} - s_{i,v,h}) - (s_{i,v,h+3} - s_{i,v,h+2}) = -s_{i,v,h+3} + s_{i,v,h+2} + s_{i,v,h+1} - s_{i,v,h} \quad (7)$$

be the overall travel time variation from $a_{i,h}$ to $a_{i,h+2}$. Thus, along the sequence $a_{i,h}$, $a_{i,h+1}$, and $a_{i,h+2}$. Also, let:

$$\Delta\tau_{h,2} = (s_{i,v,h+1} - s_{i,v,h}) - (s_{i,v,h+2} - s_{i,v,h+1}) = -s_{i,v,h+2} + 2s_{i,v,h+1} - s_{i,v,h} \quad (8)$$

be the travel time variation along the sequence of arcs $a_{i,h}$ and $a_{i,h+1}$. If the vehicle is smoothly accelerating, then $\Delta\tau_{h,3}$ and $\Delta\tau_{h,2}$ are both positive. If $\Delta\tau_{h,3} > 0$ and $\Delta\tau_{h,2} < 0$, the vehicle accelerates globally but decelerates in the middle arc. Therefore, the term:

$$|\Delta\tau_{h,3} - \Delta\tau_{h,2}| = |s_{i,v,h+3} - 2s_{i,v,h+2} + s_{i,v,h+1}| \quad (9)$$

is reduced if $\Delta\tau_{h,3}$ and $\Delta\tau_{h,2}$ are both positive or negative and, in particular, is null if $\Delta\tau_{h,3} = \Delta\tau_{h,2}$.

4.1.1. Problem constraints

The problem in (4) is subject to a set of constraints that guarantee some properties of vehicle kinematics. In particular, the set of equations:

$$s_{i,v,h+1} = t_{i,v,h}, \quad \forall v \in \mathcal{V}_i, \forall h = 1, \dots, n_i - 1, \forall i \in \mathcal{S} \quad (10)$$

guarantees the continuity of the time-space trajectories, while the set of equations:

$$s_{i,v,0} + \frac{\Delta_i}{v_{rd}^{\max}} \leq s_{i,v,1} \leq s_{i,v,0} + \frac{\Delta_i}{v_{rd}^{\min}} \quad \forall v \in \mathcal{V}_i, \forall i \in \mathcal{S} \quad (11)$$

guarantees that vehicles are not required to reach the first point of the trajectories too early or too late in compliance with the minimum speed $v_{rd}^{\min}$ and the maximum speed $v_{rd}^{\max}$ allowed in the road stretch between the triggers and $P_{i,1}$.

In addition, to avoid high, uncomfortable accelerations, the problem in (4) is subject to:

$$\begin{cases} t_{i,v,h+1} - s_{i,v,h+1} \geq t_{i,v,h} - s_{i,v,h} - \Phi_l \\ t_{i,v,h+1} - s_{i,v,h+1} \leq t_{i,v,h} - s_{i,v,h} + \Phi_l \end{cases} \quad \forall v \in \mathcal{V}_i, \forall h = 1, \dots, n_i - 1, \forall i \in \mathcal{S} \quad (12)$$

and

$$\begin{cases} t_{i,v,h+1} - s_{i,v,h+1} \geq (t_{i,v,h} - s_{i,v,h}) \cdot \left(1 - \frac{\Phi_h}{\Phi_{\min}}\right) \\ t_{i,v,h+1} - s_{i,v,h+1} \leq (t_{i,v,h} - s_{i,v,h}) \cdot \left(1 + \frac{\Phi_h}{\Phi_{\min}}\right) \end{cases} \quad \forall v \in \mathcal{V}_i, \forall h = 1, \dots, n_i - 1, \forall i \in \mathcal{S} \quad (13)$$

These sets of equations guarantee that vehicles do not vary their speed too rapidly. In particular, the constraints in (12) guarantee that if the travel time on arc $a_{i,h}$ is long (i.e., if vehicle v is traveling with low speed), the maximum reduction in the travel time (i.e., the maximum speed increase) is limited by Φ_l . In contrast, the constraints in (13) guarantee that if the travel time on arc $a_{i,h}$ is short (i.e., if vehicle v is traveling with high speed), the maximum reduction in the travel time (i.e., the maximum speed increase) is limited by $\Phi_h < \Phi_l$. This choice allows a larger acceleration when a vehicle travels slowly (i.e., a more rapid reaction of vehicles that have been slow down to give way). If $(t_{i,v,h} - s_{i,v,h})\Phi_h < \Phi_l\Phi_{\min}$ (i.e., if $t_{i,v,h} - s_{i,v,h}$ is small (i.e., high speed)), constraints (12) are dominated by constraints (13). Suitable values of Φ_l and Φ_h can be achieved by considering the continuous-time accelerations of vehicles obtained by interpolation of the solution as described in Section 4.3.

A further set of constraints is considered to guarantee a safe crossing of the intersection. In particular:

$$t_{i,v,h} - s_{i,v,h} \geq \frac{\Phi_{\min}}{r_{i,h}} = \frac{\delta}{v_{\max} \cdot r_{i,h}}, \quad \forall v \in \mathcal{V}_i, \forall h = 1, \dots, n_i, \forall i \in \mathcal{S} \quad (14)$$

guarantees that vehicles respect the maximum speed limit along the curved trajectories, with $r_{i,h}$ being the speed reduction factor, while:

$$s_{i,v+1,h} \geq t_{i,v,k} + \beta, \quad \forall \{P_{i,h}, P_{i,k}\} \in \mathcal{I}^S, \forall v \in \mathcal{V}_i \quad (15)$$

defines the minimum distance between two vehicles on the same stream. In particular, a generic follower vehicle $v+1$ cannot cross point $P_{i,h}$ until the leader vehicle v in the stream has reached point $P_{i,k+1}$, where $\{P_{i,h}, P_{i,k}\} \in \mathcal{I}^S$.

Regarding vehicles traveling along conflicting trajectories $(i, j) \in \mathcal{C}$, the set of equations:

$$\begin{cases} s_{i,v,h} \geq t_{j,u,k} + \beta - M x_{i,v,j,u} \\ s_{j,u,k} \geq t_{i,v,h} + \beta - M (1 - x_{i,v,j,u}) \end{cases} \quad \forall \{P_{i,h}, P_{j,k}\} \in \mathcal{I}^C, \forall v \in \mathcal{V}_i, u \in \mathcal{V}_j \quad (16)$$

manages the incompatibility between vehicles' relative positions; in particular, if the binary variable $x_{i,v,j,u}$ is set to 0, vehicle v in stream i cannot leave point $P_{i,h}$ until vehicle u in stream j has reached point $P_{j,k+1}$ (and after an additional headway β). However, if $x_{i,v,j,u} = 1$, then u cannot cross $P_{j,k}$ until v has reached $P_{i,h+1}$. These constraints are set for all the conflicting points $\{P_{i,h}, P_{j,k}\}$ in the incompatibility set $\mathcal{I}^C$.

Ultimately, the set of equations:

$$\begin{cases} s_{i,v,h} \geq t_{j,u,k} + \beta - M x_{i,v,j,u} \\ s_{j,u,k} \geq t_{i,v,h} + \beta - M (1 - x_{i,v,j,u}) \\ s_{i,v,0} \geq s_{j,u,0} - M x_{i,v,j,u} \\ s_{j,u,0} \geq s_{i,v,0} - M (1 - x_{i,v,j,u}) \end{cases} \quad \forall \{P_{i,h}, P_{j,k}\} \in \mathcal{I}^O, \forall v \in \mathcal{V}_i, u \in \mathcal{V}_j \quad (17)$$

manages the incompatibility between vehicles' relative positions traveling along partially overlapped trajectories $(i, j) \in \mathcal{O}$. In particular,

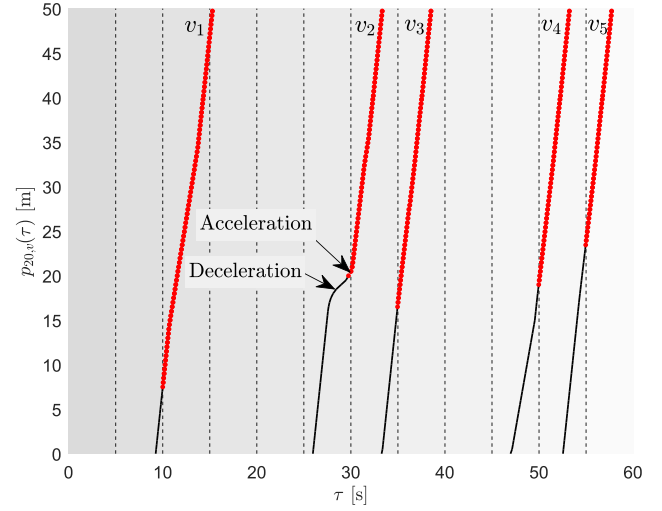


Fig. 8. Sequence of intervals $\Delta\tau = 5$ s and relevant vehicles in stream $i = 20$. The continuous black lines represent the time-space diagrams determined by the model in a certain time interval, while the dotted red lines represent their portion that, being referred to vehicles still inside the intersection at the end of the time interval, constitutes additional constraints in such an interval.

the third and the fourth equations identify which vehicle enters the intersection first, while the first and the second equations set the right of way in the conflicts as in constraints in (16).

4.2. Problem solution and the rolling horizon framework

Since the problem in (4)–(17) is computationally difficult to solve, the solution time is, in general, not compatible with real-time applications. Therefore, in the following, an approach for reducing the computational time is proposed. The pseudocode with the complete steps of this approach is shown in Alg. 1. More specifically, as mentioned in Section 3, the considered architecture assumes solving a temporal-discretized sequence of problems by considering, in each problem, only the vehicles arriving at the intersection in the interval $\Delta\tau$. Therefore, to reduce the computational complexity, it is possible to reduce the number of vehicles in each problem instance by shortening the time window $\Delta\tau$. Thus, a rolling horizon framework must be used using the following steps:

1. Consider the interval $[\tau_k, \tau_{k+1})$, $k = 1, 2, \dots$ of length $\Delta\tau$ defined in Section 3;
2. The vehicles entered in the previous time intervals that are still in the intersection in $[\tau_k, \tau_{k+1})$ are considered in the relevant optimization problem as part of constraints; to accomplish this, the set $\mathcal{V}_k^{\text{in}}$ with all the vehicles still inside the intersection is created first. Second, for each vehicle v in $\mathcal{V}_k^{\text{in}}$, the values of the variables $s_{i,v,h}$, $t_{i,v,h}$ and $x_{i,v,j,u}$, which are determined at a previous time interval, are fixed using constraints for each point $P_{i,h}$ of the trajectory γ_i followed by v ;
3. The optimization problem is solved with all the vehicles, and the solutions related to the new entering vehicles (vehicles in the set $\mathcal{V}_k$) are stored in the solution set $\mathcal{J}$;
4. The total travel time Θ related to wider intervals corresponding to the union of several time windows is the sum of the total travel times Θ_k in all the time windows, that is $\Theta = \sum_k \Theta_k$. Therefore, to avoid multiple counts of the same vehicles, in Θ_k , only the vehicles whose speed is optimized in the interval $[\tau_k, \tau_{k+1})$ are considered, while those still in the intersection but arrived in previous intervals are neglected.

Algorithm 1 Solution algorithm

```

 $\mathcal{J} \leftarrow \emptyset$ 
 $\mathcal{V}_k^{\text{in}} \leftarrow \emptyset$ 
 $\mathcal{H}_k \leftarrow \emptyset$ 
for  $k = 1, 2, \dots$  do
   $I_k \leftarrow [\tau_k, \tau_{k+1})$ 
   $\mathcal{V}_k \leftarrow \{v | v \text{ vehicles entering the intersection in } I_k\}$ 
  if  $k \geq 2$  then
     $\mathcal{V}_k^{\text{in}} \leftarrow \{v | v \text{ still inside the intersection in } I_k\}$ 
    for  $v \in \mathcal{V}_k^{\text{in}}, P_{i,h} \in P_i$  do
      find in  $\mathcal{J}$  the values of  $s_{i,v,h}$ ,  $t_{i,v,h}$ , and  $x_{i,v,j,u}$ 
      write the set of constraints  $\mathcal{H}_k$  to fix the values of the
      variables  $s_{i,v,h}$ ,  $t_{i,v,h}$ , and  $x_{i,v,j,u}$ 
    end for
  end if
  with all the vehicles in the set  $\mathcal{V}_k^{\text{in}} \cup \mathcal{V}_k$ , state and solve the
  optimization problem
   $[s^*, t^*, x^*] = \arg \min_{s,t,x} (\Theta + \alpha SF)$  s.t. constraints in Eqs.
  (10)–(17) and constraints in  $\mathcal{H}_k$ 
  for  $v \in \mathcal{V}_k$  do
    update  $\mathcal{J}$  with the solution related to  $v$ 
  end for
end for

```

A graphical example is shown in Fig. 8 and shows some optimal time–space trajectories for stream $i = 20$ over a minute assuming $\Delta\tau = 5$ s. For example, the optimal time–space diagram of vehicle v_1 is determined for interval $[5, 10)$ and constitutes a constraint in the problem of interval $[10, 15)$. In compliance with the above third step, the nonnull terms Θ_k are Θ_2 , which considers only v_1 ; Θ_6 , which considers only v_2 ; Θ_7 , which considers only v_3 ; Θ_{10} , which considers only v_4 ; and Θ_{11} , which considers only v_5 .

4.3. Continuous approximation of the solution

The proposed MILP problem provides a discrete solution, but the movement of vehicles along trajectories is continuous. Therefore, although such a solution could be sufficient for the vehicle controller, it is interesting to investigate the continuous approximation of the speed and acceleration of vehicles and analyze their properties. To achieve this goal, the MILP solution can be considered for each v to be the sequences of pairs $(P_{i,h}, s_{i,v,h}^*)$, where $P_{i,h}$ is a point of the trajectory and $s_{i,v,h}^*$ is the time at which vehicle v reaches it. Such a sequence can be rewritten as the sorted set of pairs $(\ell_{i,h}, s_{i,v,h}^*)$, which express the times $s_{i,v,h}^*$ at which vehicle v is at a distance $\ell_{i,h}$ from $P_{i,1}$.

Therefore, by interpolating all the entries of the sequence, it is possible to find a continuous function $f_{i,v}$ providing the time at which vehicle v traveling along γ_i reaches any point P of its trajectory. Because $s_{i,v,h+1}^* > s_{i,v,h}^*$, the monotonicity of $f_{i,v}$ is a requirement that can be met using a monotone cubic Hermite spline interpolation (Fritsch and Carlson, 1980). Due to the characteristics of the chosen spline form, the function $f_{i,v}$ is bijective and, denoting with $p_{i,v}(\tau)$ the distance traveled by v from $P_{i,1}$ at time τ , it is possible to find the inverse function $f_{i,v}^{-1} : p_{i,v}(\tau) = f_{i,v}^{-1}(\tau)$ expressing the distance traveled by vehicle v at τ . In particular, since $f_{i,v}$ is an interpolating function, it is guaranteed that $p_{i,v}(s_{i,v,h}) = f_{i,v}^{-1}(s_{i,v,h}) = \ell_{i,h}$. The resulting increasing monotonicity of $f_{i,v}^{-1}$ guarantees that vehicle v never stops or goes backward.

Then, it is possible to verify if the safety constraint (14) is always met:

$$\dot{p}_{i,v}(\tau) \leq v_{\text{int}}^{\max} \cdot r \left(f_{i,v}^{-1}(\tau) \right), \quad \tau \in \mathbb{R} | s_{i,v,0} \leq \tau \leq t_{i,v,n_i} \quad (18)$$

Table 3

Model parameters.

Parameter	Value
$\ell_{\text{int}}^{\max}$	10 ms ⁻¹
$\ell_{\text{rd}}^{\max}$	10 ms ⁻¹
Φ_l	0.05 s
$\Phi_{\min}$	0.05 s
Φ_h	0.025 s
Δ_j	50 m
β	0.0 s (unless otherwise stated)
α	$\alpha = \hat{\theta}/2SF$ (unless otherwise stated)

Table 4

Incoming flows for the three demand scenarios.

Lane	Lane flow [veh·h ⁻¹]			Turning rates [%]		
	Low	Medium	High	Left	Straight	Right
South L	600	720	1000	20	50	30
South R	600	720	1000	30	50	20
East L	420	560	680	50	25	25
East R	630	840	1020	42	33	25
North L	450	700	825	30	40	30
North R	450	700	825	30	40	30
West L	400	532	960	25	25	50
West R	600	798	1440	25	33	42

Analogously, it is possible to check if the module of instantaneous acceleration is within a given comfort threshold:

$$|\ddot{p}_{i,v}(\tau)| \leq a_{\max}, \quad \tau \in \mathbb{R} | s_{i,v,0} \leq \tau \leq t_{i,v,n_i} \quad (19)$$

The analysis of (19) allows to calibrate the values of the problem parameters Φ_l and Φ_h with respect to the physical parameter $a_{\max}$.

Thus, when these two constraints are not met, it is possible to manage their infeasibility by increasing the parameter β in (14) or reducing the parameters Φ_l and Φ_h in (12) and (13), respectively.

5. Case study and results

In this section, the application of the proposed MILP problem to a case study is discussed with the aim of:

1. Showing the vehicle motion planning and, in particular, how the model can safely avoid collisions between vehicles inside the intersection;
2. Discussing the total travel time Θ and the computational effort in different demand scenarios;
3. Comparing the performances of the described model with those obtainable with a first-come first-served (FCFS) control policy and with a traditional signalized intersection;
4. Discussing the effects of the smoothing filter SF ; and
5. Discussing the effect on the solution performance of the additional headway between the vehicles.

In all analyses, the intersection shown in Fig. 1 is considered with the model parameters reported in Table 3. In addition, to conduct some performance analyses, different randomly generated demand scenarios are considered, named ‘Low’, ‘Medium’, and ‘High’ in the following. Thus, vehicles are assumed to be characterized by Poissonian arrivals with a rate equal to the incoming hourly flows, with arrival speeds (at triggers) uniformly distributed in $[4, 10]$ ms⁻¹, and with lane flows and the turning rates reported in Table 4. As shown in Fig. 1, each lane is identified by a cardinal point (North, South, West and East) and by a letter addressing the Left (L) and Right (R) lanes. Given a lane, a cardinal point and a turning direction, it is possible to identify the relevant stream (e.g., vehicles turning left in the south-right lane are associated with stream 3 shown in Fig. 1).

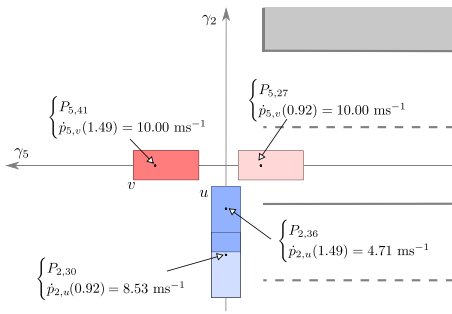


Fig. 9. First example of vehicle motion planning.

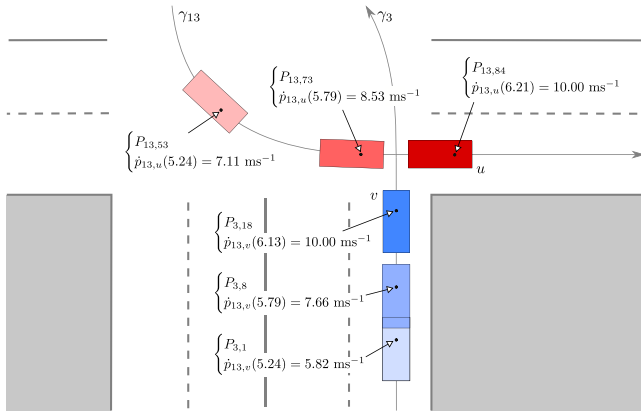


Fig. 10. Second example of vehicle motion planning: vehicle u has the right of way, and after traveling at a reduced speed along trajectory γ_{13} , it accelerates across the γ_3 area in front of vehicle v .

5.1. Vehicle motion planning and conflict management

In this section, the characteristics of feasible solutions are analyzed and discussed. To achieve this goal, the frames shown in Figs. 9 and 10 are considered, where consecutive positions of two vehicles are shown with different shades of color. In particular, the darker rectangles represent the current positions of the vehicles in the last frame, while the lighter rectangles represent the previous ones.

Looking at the two frames in Fig. 9, it is possible to note that vehicle v in stream 5 is right of way; thus, it crosses the intersection before vehicle u in stream 2, always traveling at the maximum speed. In contrast, u decelerates and reaches the point $P_{2,36}$ at $\tau = 1.49$ s, when v is already on the point $P_{5,41}$.

A second example is shown in Fig. 10, where it is possible to note that vehicle u in stream 13, traveling at a low speed along curve trajectory γ_{13} due to the vehicle stability constraint in (14), can accelerate as the trajectory curvature decreases and pass in front of vehicle v in stream 3; u has the right of way. Concurrently, vehicle v was initially traveling at a low speed to give way to u but can eventually accelerate to reach the point $P_{3,18}$ when u is on $P_{13,84}$.

5.2. Computational complexity

The goal of this section is to discuss the global performances and the computational effort in case studies characterized by a time horizon of 60 s. Thus, three solution approaches are compared:

- (a) The time horizon is solved while considering the problem with all the vehicles that arrive at the intersection in the interval $[0, 60]$;

Table 5

Comparison of the three solution approaches in terms of the average minimized total travel time Θ and the average computational time CT . Averages are computed over 10 runs.

Demand scenario		Low	Medium	High
Average number of vehicles		57	92	133
a	Θ_a	536.15 s	887.43 s	1326.92 s
	CT_a	100.94 s	11.4 h s	2 days ^a
b	Θ_b	548.98 s	896.73 s	1271.25 s
	CT_b	0.278 s	1.09 s	2.37 s
c	Θ_c	556.72 s	936.17 s	1294.19 s
	CT_c	0.40 s	0.475 s	0.67 s

^a Best solution value after two days of processing time.

- (b) The time horizon is divided into twelve time windows with $\Delta\tau = 5$ s; and

- (c) The time horizon is divided into twenty time windows with $\Delta\tau = 3$ s.

In the second and third cases, the rolling horizon framework described in Section 4.2 is used.

The solutions were computed using the solver IBM[™] Cplex[™] on an Intel[™] Xeon[™] CPU @ 3.4 GHz and 8 Gb RAM. Table 5 shows the total travel time Θ and the computational time CT for the three solution approaches and for the three demand scenarios. With approach a , the required computational time for all three demand scenarios is too high and not compatible with real world applications. In particular, the search for the solution in the high-demand scenario was stopped after two days, and the found solution is suboptimal compared to those found with approaches b and c .

In contrast, with the other approaches, the average CT is always markedly less than 5 s (b) and 3 s (c), indicating that the optimal solution can be found within the relevant $\Delta\tau$ and providing an estimate of the maximum allowed time for solution e . The CT in approach c is on average less than those in approach b , as with $\Delta\tau = 3$ s, where few vehicles are considered in any instance. However, in the scenario with low demand, $CT_b < CT_c$ due to the overhead required to ‘prepare’ and solve a higher number of subproblems, even if they are simpler.

Regarding performance, the total travel time Θ is better with the approach a , as expected, because the problem considers all the vehicles at a time, apart from the high-demand scenario where only a suboptimal solution was found. However, the differences between Θ_b and Θ_c with respect to Θ_a are negligible. Approach b provides better (lower) total travel times than approach c due to the higher number of vehicles that are considered in any instance of approach b . Therefore, a balance between the acceptable suboptimality of the solution and the relevant computational time must be chosen a priori, primarily depending on the computational resources that can be involved in the problem solution.

5.3. Simulation results and comparisons

In this section, the performance of the proposed approach is compared with the one obtained with a first-come first served (FCFS) control policy and with the one obtained with a more traditional traffic signal-based solution, which has been optimized for the same vehicle flows.

For both the three considered methods, the following performance indicators have been computed and used for the comparison:

1. Total travel time computed over all the vehicles entering the intersection area in 60 s;
2. Number of vehicles crossing the intersection in 60 s.

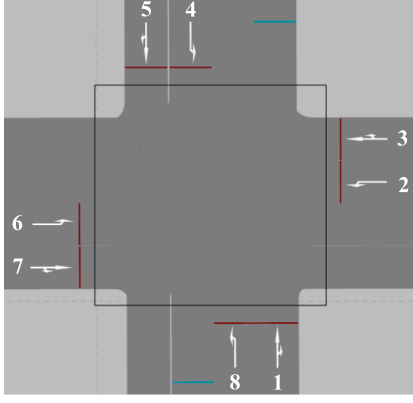


Fig. 11. Layout of the intersection considered in the simulation and movement numbers.

In all the three cases, the intersection of Fig. 1 was considered. These performance metrics are estimated for the proposed method and for the FCFS policy directly from the model output, and for the signal-based control policy through a simulation in PTV Vissim™. In all cases, 10 independent runs of 1800 s were executed, with a warm-up period of 180 s, which was sufficient according to the batch means method; for each run, the performance metrics have been computed every 60 s; the results are the averages over the runs of the averages over the simulation time. Concerning the driving behavior, the desired speed was set to 40 km/h, vehicles were not allowed to change lane in the intersection, and the Wiedemann 74 model (Wiedemann, 1974) was used to model the car-following behavior, as it is common for urban roads in Vissim™. Since it was not possible to calibrate the Wiedemann 74 parameters, average values among the most realistic and common ones in literature were used (Al-Msari et al., 2024); in particular, the standstill distance ax was set to 0.5 m, the additive factor for security distance $bxAdd$ to 1.5, and the multiplicative factor for security distance $bxMult$ to 2.5. All other simulation parameters and settings were left to their default value.

FCFS is a common control policy for AIM (Dresner and Stone, 2004; Abbas-Turki et al., 2023), based on the rule of defining the vehicle passing orders on the basis of their arrival time at intersections. The FCFS policy was simulated by adding in the proposed model the following constraint to Eq. (16):

$$\begin{cases} s_{i,v,0} \geq s_{j,u,0} - M x_{i,v,j,u} \\ s_{j,u,0} \geq s_{i,v,0} - M(1 - x_{i,v,j,u}) \end{cases} \quad \forall \{P_{i,h}, P_{j,k}\} \in \mathcal{I}^C, \forall v \in \mathcal{V}_i, u \in \mathcal{V}_j \quad (20)$$

which was used to assign vehicle right of ways (defined by the Boolean variable $x_{i,v,j,u}$) to the vehicle arrived first at the intersection. In this way, $x_{i,v,j,u}$ is no more a decision variable of the problem, but it depends on the vehicle arriving times, which are an input of the model. It is worth pointing out how the FCFS control policy can be integrated in the proposed model by simply adding Eq. (20).

Regarding the signal-based control policy, a traffic light with 8 movements and 4 phases has been considered. In a traditional intersection with a traffic light, the left lanes allow only left turnings, and the right lanes allow only going straight or turning right, as shown in Fig. 11. Concerning the traffic scenario, the incoming flows are those reported in Table 4, although for each of the four directions, they have been redistributed between the left and the right lanes to consider the new layout.

Table 6
Traffic light setup used for comparison.

Phase	Movements activated (sat. flow [veh·h ⁻¹])	Green time interval [s]		
		Low	Medium	High
1	4 (2500) – 8 (2500)	[110, 120)	[108, 120)	[0, 10)
2	2 (2500) – 6 (2500)	[46, 64)	[0, 19)	[10, 25)
3	1 (2000) – 5 (1700)	[0, 46)	[66, 108)	[25, 63)
4	3 (1800) – 7 (1600)	[64, 110)	[19, 66)	[63, 120)

Table 7
Simulation results, reported as averages over 10 runs, for the three compared methods.

Demand scenario	Total travel time [s]			Vehicle crossings		
	Traffic light	FCFS	Problem in (4)–(17)	Traffic light	FCFS	Problem in (4)–(17)
Low	2750	786	548	64	63	70
Medium	3106	1774	906	75	70	93
High	3356	2843	1242	75	74	131

The traffic light setup has been designed using the optimization model proposed in Improta and Cantarella (1984), which, starting from the definition of the movements and their incompatibilities, determines the best phase sequences and lengths, which maximize the intersection capacity. Table 6 shows the results of the optimized traffic signal, together with the saturation flow of each movement, which has been evaluated with Vissim™. The movement numbering refers to Fig. 11. Also, the cycle time was set to 120 s (a common value for intersections with this layout and traffic volume), the lost time was null, and the traffic light had only green/red lights. These last assumptions favored traffic light intersection performance.

5.3.1. Total travel time

The results in Table 7 show that the proposed method sensibly reduces the travel time inside the intersection area: the greatest advantage is in the low-demand scenario with respect to the traffic light, where the delaying effect of the traffic light is more evident, whereas the proposed method allows vehicles to cross the intersection without stops. The proposed method also performs significantly better than the FCFS policy, especially in the high-demand scenario, where the negative effect of determining the vehicle passing orders on the basis of their arrival time is more evident, due to the lower vehicle interarrival time.

5.3.2. Vehicle crossings

As for vehicle crossings, the proposed method always performs much better than the traffic light and the FCFS policy (which tends to have a performance similar to that of traffic light), particularly in the medium- and high-demand scenarios, with a particularly interesting increase ratio of around 1.8 in this last case.

5.4. Effects of the smoothing filter

In this section, the effect of the smoothing filter and the solution characteristics for different values of α are discussed. To achieve this goal, three values of such a parameter are considered: $\alpha = 0$, which represents the case without a filter; $\alpha = \hat{\theta}/2SF$ and $\alpha = \hat{\theta}/SF$, which represent the cases where SF is made comparable to half of the average total travel time and to the average total travel time, respectively. In this framework, $\hat{\theta}$ and SF are estimated by averaging the values of the cost function computed by considering only the first term and the second terms in (4), respectively.

Results are summarized in Figs. 12 and 13, where the speed of vehicles and the time-space diagrams are reported for the considered values of α . In such figures, the speeds tend to be smoother and more comfortable when α increases. If $\alpha > 0$, the number of accelerations/decelerations decreases, although they cannot be totally avoided due to

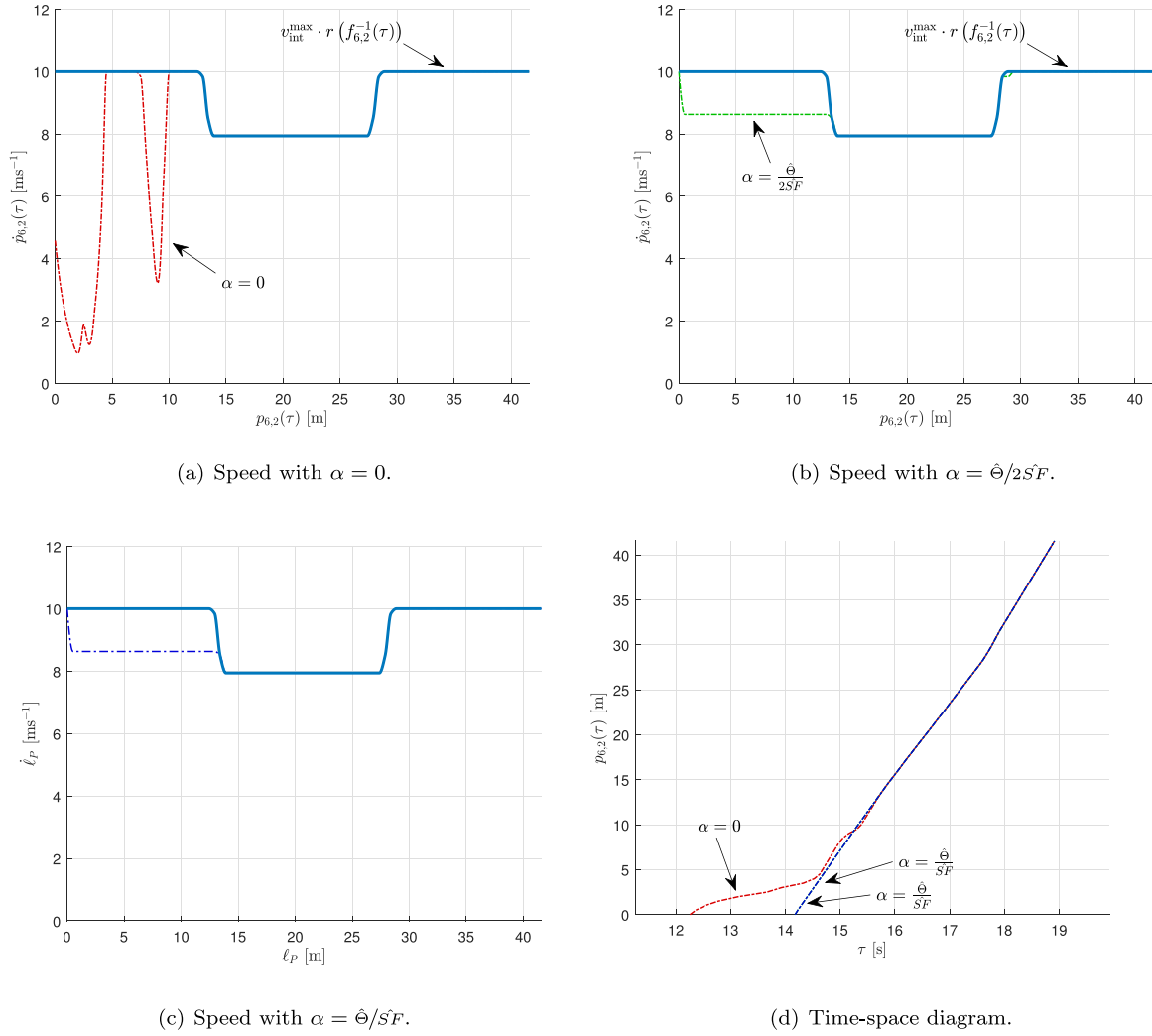


Fig. 12. Example of the optimal time-space trajectory and speed (dotted lines) of the second vehicle in stream 6. The continuous blue lines represent the speed limit. The speeds and the time-space diagrams of cases $\alpha = \hat{\Theta}/2SF$ and $\alpha = \hat{\Theta}/SF$ are indistinguishable.

Table 8

Average total travel time Θ for different values of the parameter α and for the three demand scenarios. Averages are computed over 10 runs.

Demand scenario	$\alpha = 0$	$\alpha = \hat{\Theta}/2SF$	$\alpha = \hat{\Theta}/SF$
Low	533.93 s	533.99 s	535.34 s
Medium	891.65 s	891.98 s	890.99 s
High	1357.43 s	1356.10 s	1356.53 s

traffic conditions and safety constraints. In any case, no significant differences can be identified between the cases $\alpha = \hat{\Theta}/2SF$ and $\alpha = \hat{\Theta}/SF$, demonstrating a negligible sensitivity to this parameter.

For the global effect of the smoothing filter, the results reported in Table 8 show that it negligibly affects the average Θ . Such behavior is easily explained by analyzing the time-space diagrams in Fig. 12d and Fig. 13d: with $\alpha = 0$, vehicles reach the first point $P_{i,1}$ of the trajectories with high speed and then decelerate to fulfill the conflict constraints; with $\alpha \neq 0$, vehicles reach $P_{i,1}$ with lower speed, decelerating after passing the trigger. While this second behavior allows avoiding a sudden deceleration of the vehicles inside the intersection, it does not affect the total travel time. In fact, because the travel time of a vehicle is given by the time elapsed from the passage on the trigger and the time the vehicle leaves the intersection, it is not relevant if the vehicles are delayed between the trigger and $P_{i,1}$ or between $P_{i,1}$ and P_{i,n_i} .

5.5. Sensitivity analysis with respect to the additional headway between vehicles

This last section discusses how the parameter β (additional vehicle headway) may affect the performance of the proposed approach in terms of total travel time and vehicle crossings. This parameter has the dimension of a time, and is used in Eqs. (15)–(17) to add a headway (time interval) between vehicles at each conflict (i.e., for each pair of incompatible points, it delays the crossing of the incompatibility of the second vehicle). Since vehicles cannot stop in the intersection, this results in increasing the minimum distance between vehicle of a value depending on the current speed. It is worth pointing out that the presence of this parameter is not strictly necessary, as the model already avoids conflicts between vehicles, and the pairs of incompatible points are determined (Section 3.5) in such a way to guarantee always a minimum distance between vehicles in the intersection. However, an additional headway (or vehicle spacing) could be demanded by safety standards, but could also be used to simulate the presence in the intersection of vehicles longer than the considered ones.

In particular, four values of β are considered: $\beta = 0.0$ s, which corresponds to no increase of the minimum distance; and $\beta = 0.3$ s, $\beta = 0.6$ s and $\beta = 1.0$ s, which correspond to an increase in the safety distance of, respectively, 3.0, 6.0 and 10.0 m if the vehicles travel at their maximum allowed speed.

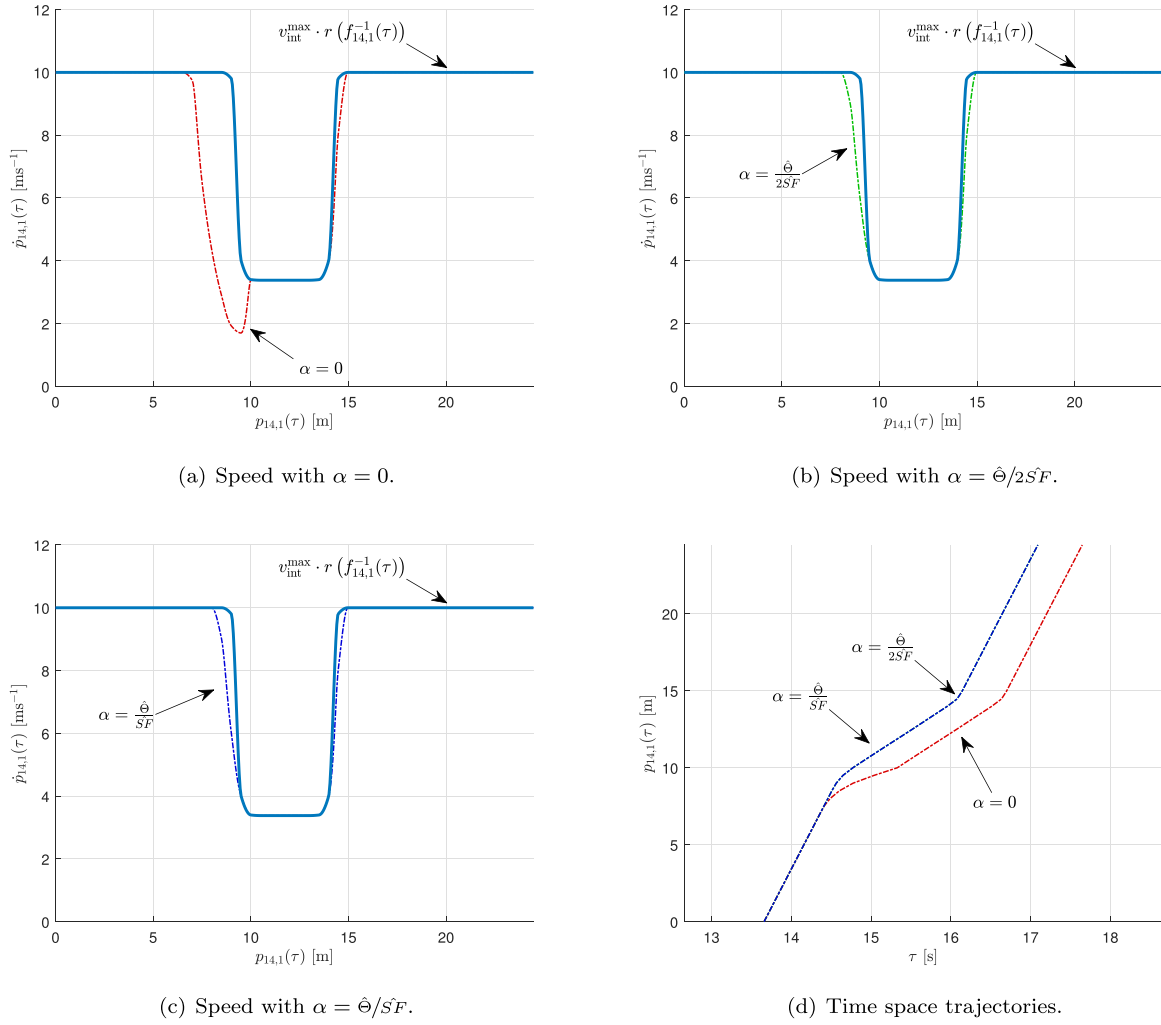


Fig. 13. Example of the optimal time-space trajectory and speed (dotted lines) of the first vehicle in stream 14. The continuous blue lines represent the speed limit. The speeds and the time-space diagrams of cases $\alpha = \hat{\theta}/2SF$ and $\alpha = \hat{\theta}/SF$ are indistinguishable.

Table 9

Average total travel time Θ and vehicle crossings (in 60 s) for different values of the parameter β and for the three demand scenarios. Averages are computed over 10 runs.

Demand scenario		Low	Medium	High
$\beta = 0.0$	Θ	549 s	897 s	1271 s
(minimum distance at $v_{\text{int}}^{\text{max}}$: 0.0 m)	vehicle crossings	70	92	133
$\beta = 0.3$	Θ	672 s	988 s	1661 s
(minimum distance at $v_{\text{int}}^{\text{max}}$: 3.0 m)	vehicle crossings	69	90	130
$\beta = 0.6$	Θ	774 s	1051 s	2095 s
(minimum distance at $v_{\text{int}}^{\text{max}}$: 6.0 m)	vehicle crossings	66	84	128
$\beta = 1.0$	Θ	933 s	1447 s	2100 s
(minimum distance at $v_{\text{int}}^{\text{max}}$: 10.0 m)	vehicle crossings	57	82	88

Table 9 shows the results for each of the three demand scenarios: as expected, increasing the minimum distance between the vehicles results in an increase in the total travel time and in a decrease in the number of vehicles that can cross the intersection in the considered time period (60 s). The decrease in the vehicle crossings is negligible for small β (e.g., $\beta \leq 0.3$) but is higher with $\beta = 1.0$, particularly in the high-demand scenario, where the number of vehicle crossings is reduced up to 34%. These results suggest that considering other typologies of vehicles (situation that could be simulated, for instance, by assigning $\beta \in (0; 0.3]$) with larger dimensions would not markedly affect the capacity of the intersection because the model is effectively able to

address larger space occupancies. Similarly, these results show how the model would be able to cope with additional safety requirements related to the vehicle spacing (and that could be achieved by assigning to β small, but nonnull, values), without having a significant worsening in the performance.

The total travel time increases with a slightly higher rate than the vehicle crossings, as a result of the increase in the value of β , especially in the high-demand scenario. For this scenario, the increase rate can be up to 70%.

6. Conclusions

In this paper, an MILP-based approach to determining the optimal scheduling and motion planning of automated vehicles (SAE levels 4 or 5) crossing intersections is proposed so that they can safely cross the intersection without the need for a traffic signal and in the minimum time. The primary advantage of the considered methodology is its ability to determine a set of suitable vehicle trajectories off-line given an intersection layout, the trajectory discretization, the computation of the incompatibility sets for conflict avoidance, and speed reduction factors along curves. Therefore, only the computation of safe and comfortable vehicle speeds that minimize the total travel time in crossing the intersection is left to online management, which is effective both in terms of performance and computational effort thanks to a rolling-horizon solution approach also introduced in the paper. In addition, the

proposed approach is sufficiently general and can potentially be applied to any type of intersection layout and vehicle type.

Compared with both a first-come first-served control policy and a signalized optimal control policy, the proposed method achieves a marked decrease in vehicle crossing times and an increase in the number of vehicle crossings.

Future work is planned to apply the proposed approach to the solution of higher-order decision problems, such as optimal vehicle routing in a network.

Another future research direction related to this field of study is the extension of the proposed model to other applications and transportation modes, such as the coordination of autonomous ships in port areas.

CRedit authorship contribution statement

Federico Gallo: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Alessandra Miagostovich:** Software, Methodology. **Davide Giglio:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Angela Di Febbraro:** Conceptualization. **Nicola Sacco:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Analytic forms of the trajectories

In this appendix, some details about the form of the trajectories and the procedure used to determine the maximum speed constraint for a general curved trajectory are reported. Consider the generic curve λ representing the trajectories shown in Fig. 3. λ consists of the following parts:

- λ_1 and λ_5 (from P_1 to A and after the point E , respectively) are the initial and final straight trajectories;
- λ_3 (from B to D) is the central circular trajectory with a fixed radius;
- λ_2 and λ_4 (from A to B and from D to E , respectively) are the clothoids that guarantee a smooth variation of the radius of curvature between λ_1 and λ_3 and between λ_3 and λ_5 .

However, for conciseness and due to the symmetry property of the considered curve trajectories, it is possible to consider only the curves λ_1 , λ_2 , and the first half of λ_3 , which are denoted as $\lambda_{3/2}$ (i.e., up to the point C). Therefore, with reference to Fig. 3, let $\tilde{\lambda} = \lambda_1 \cup \lambda_2 \cup \lambda_{3/2}$ and P be a generic point belonging to $\tilde{\lambda}$. It is possible to write a parametrization of P as:

$$P \in \lambda_1 \Leftrightarrow \begin{cases} P = P_1 + (0, \xi) \\ \xi \in [0, \xi_M] \end{cases} \quad (\text{A.1})$$

$$P \in \lambda_2 \Leftrightarrow \begin{cases} P = A + a \int_0^\zeta (\cos(\eta^2), \sin(\eta^2)) d\eta, \\ \zeta \in [0, \zeta_M] \end{cases} \quad (\text{A.2})$$

$$P \in \lambda_{3/2} \Leftrightarrow \begin{cases} P = C_o + (R \cos(\vartheta), R \sin(\vartheta)), \\ \vartheta \in [\vartheta_m, \vartheta_M] \end{cases} \quad (\text{A.3})$$

where $P_1 = (x_1, y_1)$ and $A = (x_1, \xi_M)$ are the initial and final points of λ_1 , respectively; a is the scale parameter of clothoid λ_2 ; and C_o and R are the center and radius of λ_3 , respectively. The relevant tangent and normal vectors $T(P)$ and $N(P)$ are:

$$P \in \lambda_1 \Leftrightarrow \begin{cases} T(P) = (0, 1) \\ \xi \in [0, \xi_M] \end{cases} \quad (\text{A.4})$$

$$P \in \lambda_2 \Leftrightarrow \begin{cases} T(P) = a_i (\cos(\zeta^2), \sin(\zeta^2)) \\ \zeta \in [0, \zeta_M] \end{cases} \quad (\text{A.5})$$

$$P \in \lambda_{3/2} \Leftrightarrow \begin{cases} T(P) = R_i (-\sin(\vartheta), \cos(\vartheta)) \\ \vartheta \in [\vartheta_m, \vartheta_M] \end{cases} \quad (\text{A.6})$$

and:

$$P \in \lambda_1 \Leftrightarrow \begin{cases} N(P) = (0, 0) \\ \xi \in [0, \xi_M] \end{cases} \quad (\text{A.7})$$

$$P \in \lambda_2 \Leftrightarrow \begin{cases} N(P) = 2a\zeta (-\sin(\zeta^2), \cos(\zeta^2)) \\ \zeta \in [0, \zeta_M] \end{cases} \quad (\text{A.8})$$

$$P \in \lambda_{3/2} \Leftrightarrow \begin{cases} N(P) = R (-\cos(\vartheta), -\sin(\vartheta)) \\ \vartheta \in [\vartheta_m, \vartheta_M] \end{cases} \quad (\text{A.9})$$

In these definitions, the physical meaning of the parameters ξ , ζ , and ϑ and their extreme values ξ_M , ζ_M , ϑ_m and ϑ_M are different, depend on the specific parametrization of the curves λ_1 , and are chosen to guarantee the condition $\lambda \in C^2$ class. Because the curve λ is oriented and simple, it is possible to identify any generic point P using its distance $\ell(P)$ from the point P_1 , expressed as a function of the parametrization of λ itself. Given (A.1)–(A.9), it is possible to determine the radius of λ at any point P as:

$$\rho(P) = \frac{\|N(P)\|^3}{\|T(P) \wedge N(P)\|} = \begin{cases} +\infty, & \text{if } P \in \lambda_1 \\ \frac{a}{2\zeta}, & \text{if } P \in \lambda_2 \\ R, & \text{if } P \in \lambda_{3/2} \end{cases} \quad (\text{A.10})$$

For the speed constraint, assuming that the friction force $\mu g M$ is the only relevant centripetal force $M \frac{v^2}{\rho(P)}$ acting on the vehicle with speed v at point P of a planar curve, where M is the mass of the vehicle, μ is the friction coefficient and g is the gravity acceleration, the maximum speed $\ell(P)$ that guarantees the stability of the vehicle can be determined by:

$$\ell(P) = \sqrt{\mu g \rho(P)}, \quad \forall P \in \lambda \quad (\text{A.11})$$

Also, by expressing the maximum speed as $\ell(P) = r(P) \cdot v^{\max}$, it is possible to express the speed reduction factor for each point P as:

$$r(P) = \begin{cases} 1 & \text{if } P \in \lambda_1 \\ \min \left\{ 1, \frac{\sqrt{\mu g \rho(P)}}{v^{\max}} \right\} & \text{if } P \in \lambda_2 \\ \min \left\{ 1, \frac{\sqrt{\mu g R}}{v^{\max}} \right\} & \text{if } P \in \lambda_{3/2} \end{cases} \quad (\text{A.12})$$

$\ell(P)$ is therefore a property of the curve and not dependent on the problem constraints. Thus, the speed reduction factors can be computed per trajectory in the intersection once and off-line.

References

- Abbas-Turki, A., Mualla, Y., Gaud, N., Calvaresi, D., Du, W., Lombard, A., Dridi, M., Koukam, A., 2023. Autonomous intersection management: Optimal trajectories and efficient scheduling. *Sensors* 23 (3), <http://dx.doi.org/10.3390/s23031509>.
- Al-Msari, H., Koting, S., Ahmed, A.N., El-Shafie, A., 2024. Review of driving-behaviour simulation: VISSIM and artificial intelligence approach. *Heliyon* 10 (4), <http://dx.doi.org/10.1016/j.heliyon.2024.e25936>.
- Bazzi, A., Zanella, A., Masini, B.M., 2016. A distributed virtual traffic light algorithm exploiting short range V2V communications. *Ad Hoc Networks* 49, 42–57. <http://dx.doi.org/10.1016/j.adhoc.2016.06.006>.
- Bichiou, Y., Rakha, H., 2019. Real-time optimal intersection control system for automated/cooperative vehicles. *Int. J. Transp. Sci. Technol.* 8 (1), 1–12. <http://dx.doi.org/10.1016/j.ijtst.2018.04.003>.
- Cardei, I., Mutlu, C., Cardei, M., 2024. Space-time graph path planner for unsignalized intersection management with a V2V agent coordination architecture. *Theoret. Comput. Sci.* 1020, 114871. <http://dx.doi.org/10.1016/j.tcs.2024.114871>.
- Chalaki, B., Malikopoulos, A.A., 2022. Optimal control of connected and automated vehicles at multiple adjacent intersections. *IEEE Trans. Control Syst. Technol.* 30 (3), 972–984. <http://dx.doi.org/10.1109/TCST.2021.3082306>.

- Chen, L., Englund, C., 2016. Cooperative intersection management: A survey. *IEEE Trans. Intell. Transp. Syst.* 17 (2), 570–586. <http://dx.doi.org/10.1109/TITS.2015.2471812>.
- Chen, X., Sun, Y., Ou, Y., Zheng, X., Wang, Z., Li, M., 2020. A conflict decision model based on game theory for intelligent vehicles at urban unsignalized intersections. *IEEE Access* 8, 189546–189555. <http://dx.doi.org/10.1109/ACCESS.2020.3031674>.
- Choi, E.H., 2010. *Crash Factors in Intersection-Related Crashes: An On-Scene Perspective*. Technical Report, NHTSA's National Center for Statistics and Analysis.
- Cruz-Piris, L., Lopez-Carmona, M., Marsa-Maestre, I., 2019. Automated optimization of intersections using a genetic algorithm. *IEEE Access* 7, 15452–15468. <http://dx.doi.org/10.1109/ACCESS.2019.2895370>.
- Di Febbraro, A., Gallo, F., Giglio, D., Sacco, N., 2020. Traffic management system for smart road networks reserved for self-driving cars. *IET Intell. Transp. Syst.* 14 (9), 1013–1024. <http://dx.doi.org/10.1049/iet-its.2019.0675>.
- Dresner, K., Stone, P., 2004. Multiagent traffic management: a reservation-based intersection control mechanism. In: *Proceedings of the Third International Joint Conference on Autonomous Agents and Multiagent Systems*, 2004. AAMAS 2004.. pp. 530–537.
- Fayazi, S., Vahidi, A., 2018. Mixed-integer linear programming for optimal scheduling of autonomous vehicle intersection crossing. *IEEE Trans. Intell. Veh.* 3 (3), 287–299. <http://dx.doi.org/10.1109/ITV.2018.2843163>.
- Feng, Y., He, D., Guan, Y., 2019. Composite platoon trajectory planning strategy for intersection throughput maximization. *IEEE Trans. Veh. Technol.* 68 (7), 6305–6319. <http://dx.doi.org/10.1109/TVT.2019.2914163>.
- Findley, D.J., 2015. *Traffic engineering studies*. In: *Traffic Engineering Handbook*. John Wiley & Sons, Ltd, pp. 109–148.
- Fritsch, F.N., Carlson, R.E., 1980. Monotone piecewise cubic interpolation. *SIAM J. Numer. Anal.* 17, <http://dx.doi.org/10.2307/2156610>.
- Ge, Q., Sun, Q., Wang, Z., Li, S.E., Gu, Z., Zheng, S., Liao, L., 2021. Real-time coordination of connected vehicles at intersections using graphical mixed integer optimization. *IET Intell. Transp. Syst.* 15 (6), 795–807. <http://dx.doi.org/10.1049/itr2.12061>.
- Gu, Y., Wang, L., Cheng, L., Gu, G., 2023. An IoT-based framework for motion planning of connected automated vehicles at signal-free traffic intersections. *IEEE Internet Things J.* 11 (22), 35752–35761. <http://dx.doi.org/10.1109/JIOT.2023.3339194>.
- Guan, Y., Ren, Y., Li, S.E., Sun, Q., Luo, L., Li, K., 2020. Centralized cooperation for connected and automated vehicles at intersections by proximal policy optimization. *IEEE Trans. Veh. Technol.* 69 (11), 12597–12608. <http://dx.doi.org/10.1109/TVT.2020.3026111>.
- Guler, S.I., Menendez, M., Meier, L., 2014. Using connected vehicle technology to improve the efficiency of intersections. *Transp. Res. Part C: Emerg. Technol.* 46, 121–131. <http://dx.doi.org/10.1016/j.trc.2014.05.008>.
- Hadjigeorgiou, A., Timotheou, S., 2024. Optimizing the crossing sequence in autonomous intersection management with travel time and energy considerations. In: *2024 American Control Conference. ACC, IEEE*, pp. 1474–1479. <http://dx.doi.org/10.23919/ACC60939.2024.10644280>.
- Improta, G., Cantarella, G., 1984. Control system design for an individual signalized junction. *Transp. Res.* 18 (2), 147–167.
- Kamal, M.A.S., i. Imura, J., Hayakawa, T., Ohata, A., Aihara, K., 2015. A vehicle-intersection coordination scheme for smooth flows of traffic without using traffic lights. *IEEE Trans. Intell. Transp. Syst.* 16 (3), 1136–1147. <http://dx.doi.org/10.1109/TITS.2014.2354380>.
- Kamal, M.A.S., Imura, J., Ohata, A., Hayakawa, T., Aihara, K., 2013. Coordination of automated vehicles at a traffic-lightless intersection. In: *16th International IEEE Conference on Intelligent Transportation Systems (ITSC 2013)*. pp. 922–927. <http://dx.doi.org/10.1109/ITSC.2013.6728350>.
- Karthikeyan, P., Chen, W.L., Hsiung, P.-A., 2022. Autonomous intersection management by using reinforcement learning. *Algorithms* 15 (9), 326. <http://dx.doi.org/10.3390/a15090326>.
- Kloock, M., Scheffe, P., Marquardt, S., Maczjewski, J., Alrifae, B., Kowalewski, S., 2019. Distributed model predictive intersection control of multiple vehicles. In: *2019 IEEE Intelligent Transportation Systems Conference. ITSC*, pp. 1735–1740. <http://dx.doi.org/10.1109/ITSC.2019.8917117>.
- Levin, M.W., Rey, D., 2017. Conflict-point formulation of intersection control for autonomous vehicles. *Transp. Res. Part C: Emerg. Technol.* 85, 528–547. <http://dx.doi.org/10.1016/j.trc.2017.09.025>.
- Li, Z., Eleftheriadou, L., Ranka, S., 2014b. Signal control optimization for automated vehicles at isolated signalized intersections. *Transp. Res. Part C: Emerg. Technol.* 49, 1–18. <http://dx.doi.org/10.1016/j.trc.2014.10.001>.
- Li, L., Wen, D., Yao, D., 2014a. A survey of traffic control with vehicular communications. *IEEE Trans. Intell. Transp. Syst.* 15 (1), 425–432. <http://dx.doi.org/10.1109/TITS.2013.2277737>.
- Liu, Y., Long, K., Wu, W., 2023. A route planning and traffic control model for autonomous intersections by minimizing conflict zones. *IEEE Access* 11, 113678–113693. <http://dx.doi.org/10.1109/ACCESS.2023.3323405>.
- Liu, B., Shi, Q., Song, Z., El Kamel, A., 2019. Trajectory planning for autonomous intersection management of connected vehicles. *Simul. Model. Pr. Theory* 90, 16–30. <http://dx.doi.org/10.1016/j.simpat.2018.10.002>.
- Liu, C., Zhang, Y., Zhang, T., Wu, X., Gao, L., Zhang, Q., 2020. High throughput vehicle coordination strategies at road intersections. *IEEE Trans. Veh. Technol.* 69 (12), 14341–14354. <http://dx.doi.org/10.1109/TVT.2020.3029933>.
- Ma, M., Li, Z., 2021. A time-independent trajectory optimization approach for connected and autonomous vehicles under reservation-based intersection control. *Transp. Res. Interdiscip. Perspect.* 9, 100312. <http://dx.doi.org/10.1016/j.trip.2021.100312>.
- Malikopoulos, A.A., Cassandras, C.G., Zhang, Y.J., 2018. A decentralized energy-optimal control framework for connected automated vehicles at signal-free intersections. *Automatica* 93, 244–256. <http://dx.doi.org/10.1016/j.automatica.2018.03.056>.
- Martelli, M., Virdis, A., Gotta, A., Cassarà, P., Di Summa, M., 2021. An outlook on the future marine traffic management system for autonomous ships. *IEEE Access* 9, 157316–157328. <http://dx.doi.org/10.1109/ACCESS.2021.3130741>.
- Müller, E.R., Carlson, R.C., Junior, W.K., 2016. Intersection control for automated vehicles with MILP. *IFAC-Pap.* 49 (3), 37–42. <http://dx.doi.org/10.1016/j.ifacol.2016.07.007>.
- Pompigna, A., Mauro, R., 2022. Smart roads: A state of the art of highways innovations in the smart age. *Eng. Sci. Technol. an Int. J.* 25, 100986. <http://dx.doi.org/10.1016/j.jestech.2021.04.005>.
- Rodrigues de Campos, G., Falcone, P., Hult, R., Wymeersch, H., Sjöberg, J., 2017. Traffic coordination at road intersections: Autonomous decision-making algorithms using model-based heuristics. *IEEE Intell. Transp. Syst. Mag.* 9 (1), 8–21. <http://dx.doi.org/10.1109/ITSM.2016.2630585>.
- SAE International, 2021. *Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles*.
- Saías-Vázquez, J., Ordóñez Morales, E., López-Nores, M., Blanco-Fernández, Y., Bravo-Torres, J., Pazos-Arias, J., Gil-Solla, A., Ramos-Cabrera, M., 2018. Intersection intelligence: Supporting urban platooning with virtual traffic lights over virtualized intersection-based routing. *Sensors (Switzerland)* 18 (11), <http://dx.doi.org/10.3390/s18114054>.
- Shladover, S.E., 2021. Opportunities and challenges in cooperative road vehicle automation. *IEEE Open J. Intell. Transp. Syst.* 2, 216–224. <http://dx.doi.org/10.1109/OJITS.2021.3099976>.
- Tallapragada, P., Cortés, J., 2015. Coordinated intersection traffic management. *IFAC-Pap.* 48 (22), 233–239. <http://dx.doi.org/10.1016/j.ifacol.2015.10.336>, 5th IFAC Workshop on Distributed Estimation and Control in Networked Systems NecSys 2015.
- Wiedemann, R., 1974. *Simulation des straßenverkehrsflusses*. Inst. Transp. Sci. Univ. Karlsru. Ger. Heft 8.
- Wu, W., Liu, Y., Liu, W., Zhang, F., Dixit, V., Waller, S.T., 2022. Autonomous intersection management for connected and automated vehicles: A lane-based method. *IEEE Trans. Intell. Transp. Syst.* 23 (9), 15091–15106. <http://dx.doi.org/10.1109/TITS.2021.3136910>.
- Wu, J., Qu, X., 2022. Intersection control with connected and automated vehicles: A review. *J. Intell. Connect. Veh.* 5 (3), 260–269. <http://dx.doi.org/10.1108/JICV-06-2022-0023>.
- Wu, J., Yan, F., Liu, J., 2018. Effectiveness proving and control of platoon-based vehicular cyber-physical systems. *IEEE Access* 6, 21140–21151. <http://dx.doi.org/10.1109/ACCESS.2018.2800404>.
- Xu, H., Cassandras, C.G., Li, L., Zhang, Y., 2022. Comparison of cooperative driving strategies for CAVs at signal-free intersections. *IEEE Trans. Intell. Transp. Syst.* 23 (7), 7614–7627. <http://dx.doi.org/10.1109/TITS.2021.3071456>.
- Yang, K., Guler, S.I., Menendez, M., 2016. Isolated intersection control for various levels of vehicle technology: Conventional, connected, and automated vehicles. *Transp. Res. Part C: Emerg. Technol.* 72, 109–129. <http://dx.doi.org/10.1016/j.trc.2016.08.009>.
- Yang, B., Monterola, C., 2016. Efficient intersection control for minimally guided vehicles: A self-organised and decentralised approach. *Transp. Res. Part C: Emerg. Technol.* 72, 283–305. <http://dx.doi.org/10.1016/j.trc.2016.10.004>.
- Yao, Z., Jiang, H., Cheng, Y., Jiang, Y., Ran, B., 2022. Integrated schedule and trajectory optimization for connected automated vehicles in a conflict zone. *IEEE Trans. Intell. Transp. Syst.* 23 (3), 1841–1851. <http://dx.doi.org/10.1109/TITS.2020.3027731>.
- Yao, Z., Jiang, H., Jiang, Y., Ran, B., 2023. A two-stage optimization method for schedule and trajectory of CAVs at an isolated autonomous intersection. *IEEE Trans. Intell. Transp. Syst.* 24 (3), 3263–3281. <http://dx.doi.org/10.1109/TITS.2022.3230682>.
- Yu, C., Sun, W., Liu, H.X., Yang, X., 2019. Managing connected and automated vehicles at isolated intersections: From reservation- to optimization-based methods. *Transp. Res. Part B: Methodol.* 122, 416–435. <http://dx.doi.org/10.1016/j.trb.2019.03.002>.
- Zhang, K., Zhang, D., de La Fortelle, A., Wu, X., Grégoire, J., 2015. State-driven priority scheduling mechanisms for driverless vehicles approaching intersections. *IEEE Trans. Intell. Transp. Syst.* 16 (5), 2487–2500. <http://dx.doi.org/10.1109/TITS.2015.2411619>.
- Zhu, F., Ukkusuri, S.V., 2015. A linear programming formulation for autonomous intersection control within a dynamic traffic assignment and connected vehicle environment. *Transp. Res. Part C: Emerg. Technol.* 55, 363–378. <http://dx.doi.org/10.1016/j.trc.2015.01.006>.